Observer-Patch Holography and the Dark Matter Phenomenon

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Abstract

In OPH, stable gravity is the bookkeeping of agreement between finite observer patches. Imperfect agreement leaves a record-repair remainder on overlap collars. On the declared continuation packet, a localized modular-energy remainder with rank-two reconstruction, conservation, normalization, and coupling receipts contributes to the gravitational stress seen by the effective geometry. Raw scalar collar CMI remains a recovery diagnostic and is not that stress.

This paper develops that conditional idea as a dark-sector continuation. In old, settled, low-acceleration galaxies the declared response ansatz gives a MOND-like galaxy law and the baryonic Tully-Fisher scaling. In clusters and cosmology the candidate source has to move as transported stress, with its own relaxation, lensing, growth, and likelihood tests. The first audited public comparisons find the fixed conditional $\mathbb{Z}_6$ scale essentially on the aggregate RAR optimum and the slope-four BTFR limit compatible with the error-aware public fit, while the fixed BTFR normalization remains under systematic-sensitive pressure. Universal Solar-System extrapolation of the same static equation is strongly excluded by Cassini, making a source-derived applicability theorem an explicit requirement of the settled-galaxy branch.

What This Paper Contributes

The dark matter evidence is not one observation. Galaxy rotation curves, the baryonic Tully-Fisher relation, the radial acceleration relation, cluster lensing, CMB peaks, BAO, growth, and S_8 all ask for a gravitating sector that ordinary luminous matter does not supply. This paper treats the MOND-like galaxy law as a clue about equilibrium response and the ΛCDM evidence as a clue about transported stress.

The OPH move is to test whether that stress can have an observer-patch origin. Imperfect record repair leaves a collar remainder. A remainder enters the effective stress bookkeeping only after the source-localization, matching, tensor, conservation, normalization, and coupling receipts declared below; raw CMI alone does not gravitate. On the continuation branch the source has no luminous charge, and the settled-galaxy response ansatz is RAR-shaped. In clusters and cosmology the candidate becomes a transported component with its own relaxation and likelihood contracts. The paper is a continuation surface; the source packet and joint fit are decisive gates.

In the forcing chain of `../CONSISTENCY_STACK.md`, this continuation sits at the junction of C1 and C5. C1 (self-reading) requires internal record-keeping with a repair mechanism; C5 (entropic-gravitational consistency) recovers the Einstein branch from patchwise horizon thermodynamics of the record surfaces. Imperfect repair is the residue of the consistency requirement that keeps records readable across the gravitational carrier: a repair remainder that carries stress enters the same bookkeeping that C5 makes gravitational, subject to the receipts declared below. The dark/anomaly stress candidate of this paper is that residue.

Contents

1 Plain Picture

A galaxy does two jobs in this model. Its baryons source the ordinary weak field, and its finite observer-overlap collars carry a record-repair load. The load is a recovery diagnostic until the declared source packet turns localized modular-energy readings into a conserved, coupled tensor. On that conditional branch it is dark to electromagnetic probes because ordinary luminous charges are absent, and it enters the effective stress bookkeeping. Scalar CMI by itself does neither.

The RAR shape has a simple origin. The relevant repair opportunities live on cuts, so their broad count scales like $r_M/r = \sqrt{g_b/a_{0,\mathrm{OPH}}}$. Independent local opportunity counts give a Poisson zero-count law. The active fraction is $1-\exp[-\lambda_{\mathrm{collar}}\sqrt{g_b/a_{0,\mathrm{OPH}}}]$. The conservative field equation promotes that scalar response into a potential equation. The simple algebraic acceleration law is exact only when symmetry removes the curl term.

2 Three Claim Layers

The first layer is the static galaxy law. It contains the RAR shape, the BTFR limit, the effective dark density, the no-slip lensing rule, and the $\mathbb{Z}_6$/Poisson coefficient.

The second layer is the dynamic dark component. It treats the anomaly as a transported stress that relaxes toward the static equilibrium over τ_{rec}. Cluster offsets use this layer.

The third layer is cosmological dark matter. It uses the same stress in linear perturbation theory, with a background abundance, a relaxation rate, and an equilibrium kernel. CMB, BAO, lensing, growth, and S_8 test this layer.

3 Finite Covariant Parent Contract

The cosmology layer needs more than scalar rows. The object consumed by a Boltzmann solver must be a finite covariant parent from which stress, exchange, response, and perturbation variables are calculated. At regulator r, write

$$\mathfrak{P}_r = (\mathcal{X}_r, g_r, e_r, \Pi_r, Z_r, f_r, \mathsf{T}_r, \mathsf{C}_r, \mathsf{M}_r, \mathsf{R}_r).$$

Here $\mathcal{X}_r$ is a finite spacetime or cell complex, g_r the regulated metric, e_r tetrads, Π_r parallel transport between adjacent cells, $Z_r = Z_{A,r} \sqcup Z_{R,r}$ anomaly and recipient packet states, $f_{I,z} \geq 0$ packet occupation, T_r finite covariant transport, C_r reaction or repair, M_r the moment map to currents and stress, and R_r the reconstruction/refinement comparison map. A packet $z = (I, x, p, \lambda)$ must satisfy

$$g_{\mu\nu}(x)p^\mu p^\nu = -m_I^2, \qquad p^0 > 0,$$

with declared finite invariant phase-space weights w_z. For anomaly packets the Standard Model currents vanish:

$$q_z^{\mathrm{EM}} = q_z^{\mathrm{color}} = q_z^{\mathrm{weak}} = 0,$$

or a separate neutral-commutant theorem must prove that the anomaly algebra induces no Standard Model current.

The finite moments are

$$J^\mu_{I,r}(x) = \frac{1}{V_x}\sum_{z:x_z=x} w_z f_{I,z} p_z^\mu, \qquad T^{\mu\nu}_{I,r}(x) = \frac{1}{V_x}\sum_{z:x_z=x} w_z f_{I,z} p_z^\mu p_z^\nu.$$

If the anomaly stress cannot be represented by positive on-shell packets, the parent must declare an internal packet stress $\Sigma_z^{\mu\nu}$ and replace $p^\mu p^\nu$ by $p^\mu p^\nu + \Sigma_z^{\mu\nu}$. Thus the paper allows three parent branches: positive kinetic parent, kinetic parent with internal stress, or covariant action parent. It does not assume every desired effective stress is pressureless positive dust.

On the positive kinetic branch, $w_z f_{I,z} \geq 0$ and future causal momenta imply $T_{ab}t^a t^b \geq 0$ for every future timelike t^a. This is the packet-parent source of weak-energy positivity; it must not be manufactured by taking absolute values of scalar source rows.

Finite total-stress closure. Assume finite covariant summation by parts, equal-and-opposite face fluxes after parallel transport, reaction-channel four-momentum conservation

$$\sum_{I,z} \nu_{\alpha,I,z} p_z^\mu = 0,$$

and moment-map compatibility with the discrete connection. Multiplying the finite kinetic equation by $w_z p_z^\nu$ and summing gives

$$\mathrm{Div}_r\, T_{I,r}^{\mu\nu} = Q_{I,r}^\nu, \qquad Q_{I,r}^\nu(x) = \frac{1}{V_x} \sum_{z:x_z=x} w_z p_z^\nu \mathsf{C}_{I,r,z}[f].$$

Summing over sectors cancels all reaction channels, so

$$\mathrm{Div}_r \sum_I T_{I,r}^{\mu\nu} = 0.$$

Continuum promotion additionally requires

$$\|\mathsf{R}_r(\mathrm{Div}_r\, T_r) - \nabla_\mu \mathsf{R}_r(T_r)^{\mu\nu}\|_{\mathcal{W}} \le \epsilon_{\mathrm{div}}(r), \qquad \epsilon_{\mathrm{div}}(r) \to 0.$$

FLRW reduction and exchange. If the homogeneous packet distribution is invariant under a spatial group with no invariant vector and no invariant traceless symmetric rank-two tensor, then

$$\bar{q}_A^\mu = 0, \qquad \bar{\pi}_A^{\mu\nu} = 0, \qquad \bar{T}_A{}^\mu{}_\nu = \mathrm{diag}(-\bar{\rho}_A, \bar{p}_A, \bar{p}_A, \bar{p}_A).$$

Reserve Q_A^μ for exchange four-current:

$$Q_A^\mu = Q_A u^\mu + F_A^\mu, \qquad u_\mu F_A^\mu = 0, \qquad \sum_I Q_I^\mu = 0.$$

The conserved homogeneous anomaly load should be denoted separately, for example $\mathcal{N}_A = a^3 \bar{\rho}_A V_{\mathrm{com}}$.

Retarded response. Linearizing the parent around the FLRW background gives

$$\delta x' = A(k,\eta)\delta x + B(k,\eta)\delta n_b^{(A)}, \qquad \delta y_A = C(k,\eta)\delta x + D(k,\eta)\delta n_b^{(A)},$$

where $n_b^{(A)} = -u_{A\mu} J_b^\mu$. With $\partial_\eta U(\eta,\eta') = A(\eta)U(\eta,\eta')$,

$$K_A(\eta,\eta') = C(\eta)U(\eta,\eta')B(\eta') + D(\eta)\delta(\eta-\eta')$$

is the actual retarded response kernel. A scalar $B_A(k,a)$ and a scalar $\Gamma_{\mathrm{rec}}(k,a)$ are controlled reductions of this realization, not primitive inputs. If there exists $H > 0$ and $\gamma > 0$ with

$$HA + A^\dagger H \preceq -2\gamma H,$$

then $\|U(\eta,\eta')\|_H \le e^{-\gamma(\eta-\eta')}$. A one-pole relaxation rate is valid only when $\|K_A - K_A^{(1)}\|_{\mathcal{W}} \le \epsilon_{\mathrm{one\ pole}}$ in a predeclared norm.

Abundance selector boundary. The conservation equation

$$\bar{\rho}'_A + 3\mathcal{H}\bar{\rho}_A = 0$$

implies $\bar{\rho}_A = C_A a^{-3}$, but it does not determine C_A. The source-only OPH abundance requires the finite selector

$$\mathcal{E}_{A,r} = F_{A,r}(\text{quotient ensemble}, P_\star, N_{\text{CRC}}, \text{release state}, \mathfrak{P}_r), \qquad \bar{\rho}_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\text{com}}},$$

with refinement compatibility. Section 17.15 defines this selector. Promotion to a source-only abundance requires `ANOMALY_ABUNDANCE_SOURCE_RECEIPT`; without that receipt, cosmological abundance is a conditional source state rather than a source-only OPH prediction.

Receipt	Claim label
`PACKET_MASS_SHELL_RECEIPT`	finite-parent evidence gate
`TOTAL_STRESS_CLOSURE_RECEIPT`	finite-parent evidence gate
`RETARDED_RESPONSE_RECEIPT`	response-realization gate
`B_A_KERNEL_RECEIPT`	conditional on response and mode calibration
`ANOMALY_ABUNDANCE_SOURCE_RECEIPT`	source-side finite selector gate

Canonical finite covariant collar-packet parent. For a regulator r and background (X, g), the source object is

$$\mathcal{P}_r[X, g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r).$$

Here $\mathcal{C}_r$ is a finite causal cell complex with oriented faces, cell four-volumes, causal adjacency, and finite parallel transports $U_{F\to c}$; e_r is the tetrad/local-frame data; $Z_r = \bigsqcup_s Z_{s,r}$ is the packet state space for anomaly, recipient, radiation, standard matter, and any interaction sector; ω_r, p_r, f_r are invariant weights, local momenta, and occupations; Φ_r is the face flux; $\mathcal{R}_r$ is the reaction channel list; $\mathcal{G}_r$ is the finite gauge/quotient data; π_r is the restriction/refinement map; and Read_r is the local observable readout. With signature $(-+++)$, every packet used in a kinetic stress branch must satisfy $g_{ab}p_z^a p_z^b = -m_z^2$, $p_z^0 > 0$, and $\omega_z f_{c,z} \geq 0$, with the invariant measure convention declared explicitly.

Scalar-to-stress nonuniqueness. A scalar collar row, even a finite row for $\rho_A(a)$, $\rho_{A,\text{eq}}(a)$, or $B_A(k, a)$, does not determine a stress tensor. Many inequivalent packet parents can have the same scalar density while differing in pressure, sound speed, anisotropic stress, exchange current, gauge-invariant perturbation variables, and causal response. Therefore no evidence bundle may promote scalar rows to physical CMB, lensing, growth, or S_8 inputs unless the parent supplies the missing stress and response variables.

Finite MaxEnt lift. When only finitely many one-cell source readings $e_x(u_i) = T^A_{ab}(x)u_i^a u_i^b$ are available, a stress lift is not unique until the branch supplies a finite selection rule. The admissible MaxEnt lift minimizes no hidden observational residual: it maximizes the declared packet entropy subject to the source readings, mass shells, conserved charges, local frame covariance, and refinement restrictions. The lift is physical only when held-out timelike probes pass a quadraticity/tomography test and the variational stress agrees with the packet-moment stress in the predeclared norm.

Packet stress and exchange closure. For a kinetic parent,

$$T^{ab}_{I,r}(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p^a_z p^b_z, \qquad J^a_{I,r}(c) = \frac{1}{V_c} \sum_{z \in Z_{I,r}(c)} \omega_z f_{c,z} p^a_z.$$

More general packet parents may add an internal stress Σ^{ab}_z, but then Σ^{ab}_z is part of the primitive parent data and must obey the same local-frame and refinement checks. The finite transport/reaction equation must imply

$$\mathrm{Div}_r\, T^{ab}_{I,r} = Q^b_{I,r}, \qquad \sum_I Q^b_{I,r} = 0.$$

If the repair exchange is nonzero, a row labelled "recipient" is not enough: the recipient stress T^{ab}_R, exchange current Q^b_R, and closure residual $Q^b_A + Q^b_R$ must be emitted and certified. If exchange is off, the parent instead emits a repair-exchange-off receipt.

Response, rate, and gauge receipts. The finite causal response must provide a finite domain of dependence, subluminal characteristic bound, retarded support residual, and response stability residual. The transition-matrix diagnostic may be called $\gamma_{\text{repair step}}$ until the active fiber, conserved-sector decomposition, physical clock $\Delta\tau_{\text{physical}}$, and common parent response pole are certified. Only then may one write

$$\Gamma_{\text{rec}} = -\,\mathrm{Re}\,\lambda_{\text{active}}/\Delta\tau_{\text{physical}}.$$

Gauge checks must be stated in gauge-invariant or rest-frame variables; raw Newtonian/synchronous agreement is a diagnostic, not a gauge-independence receipt.

Receipt hierarchy. The finite parent receipt is the conjunction of primitive evidence receipts, not a producer-declared boolean. The source-side conjunction includes

SOURCE_ROUTE, ENTROPY_UNIT, MODULAR_NONADDITIVITY_IDENTITY, SOURCE_LOCALIZATION, STRESS
FINITE_PACKET_KINEMATICS, PACKET_MASS_SHELL, TRANSPORT_COVARIANCE, CHANNEL_FOUR_MOMENTUM, FINITE_
VARIATIONAL_MOMENT_STRESS_AGREEMENT, LOCAL_FRAME_COVARIANCE, CARRIER_QUOTIENT_INV
TOTAL_STRESS_CLOSURE, EXCHANGE_CURRENT_CLOSURE, COSMOLOGICAL_GAUGE_INVARIANC
FINITE_DOMAIN_OF_DEPENDENCE, SUBLUMINAL_CHARACTERISTICS, RETARDED_RESPONSE, RESPONSE
REFINEMENT_CONVERGENCE, CDM_LIMIT_RECOVERY.

Frozen source, solver, likelihood, and data hashes are later promotion receipts. They do not belong to the parent theorem and cannot rescue a failed parent.

4 Source Ontology, Units, and Matching

This paper uses metric signature $(-+++)$. Entropies are in nats unless a receipt explicitly declares bits and multiplies by $\ln 2$. The radius ℓ in the local source formula is the proper geodesic radius of a small four-dimensional causal diamond or ball read in the local tetrad, with scale hierarchy

$$\ell_{\text{UV}} \ll \delta_{\text{collar}} \ll \ell \ll L_{\text{curvature}}, L_{\text{gradient}}.$$

Natural units may set $\hbar = c = 1$. In SI units a dimensionless entropy source R has energy-density normalization

$$\rho_A c^2 = \frac{15\hbar c}{8\pi^2 \ell^4} R.$$

4.1 Three Source Objects

The dark continuation distinguishes

$$R_C^{\mathrm{info}} := I_\rho(A:D|B), \qquad \mathcal{M}_C^{\mathrm{anom}}, \qquad T_A^{ab}.$$

The first is raw finite conditional mutual information. The second is a source variable proved, or explicitly hypothesized on the continuation branch, to equal the anomalous local modular energy entering the BW ball calculation. The third is a tensor stress reconstructed from all-frame source readings and a finite packet parent. The paper does not identify these three objects by notation alone.

Exact finite modular-nonadditivity identity. For a faithful finite-dimensional state ρ_{ABD}, define support modular Hamiltonians $K_X = -\log \rho_X$, embedded in the full algebra, and

$$\Delta K_\rho = K_{AB} + K_{BD} - K_B - K_{ABD}.$$

Then

$$\mathrm{Tr}\, \rho_{ABD} \Delta K_\rho = S(AB) + S(BD) - S(B) - S(ABD) = I_\rho(A:D|B).$$

This is an expectation identity. It does not make ΔK_ρ a positive operator, a state-independent geometric generator, or a local collar stress.

First-variation obstruction. If σ is full rank with $I_\sigma(A:D|B) = 0$ and $\rho(t) = \sigma + tX + O(t^2)$, then

$$\frac{\mathrm{d}}{\mathrm{d}t} I_{\rho(t)}(A:D|B)\bigg|_{t=0} = 0$$

because the nonnegative CMI functional has a local minimum on the exact Markov set. A fixed-reference modular-energy variation $\delta\langle K_\sigma\rangle = t\ \mathrm{Tr}(XK_\sigma) + O(t^2)$ is generically linear. Therefore raw CMI is not generically the same object as the anomalous local modular energy on a perturbative Markov branch.

4.2 Declared Source Route

A source artifact must declare exactly one route:

$$\texttt{source\_route} \in \{\texttt{FIXED\_REFERENCE\_MODULAR\_ENERGY}, \texttt{NONLINEAR\_CMI\_STRESS}\}.$$

There is no automatic mixed mode. On the fixed-reference route the source is

$$R_C^{\mathrm{src}} = \mathcal{M}_C^{\mathrm{anom}} = \delta\langle K_C^{(\mathrm{anom})}[\sigma]\rangle,$$

and CMI is a recovery/localization diagnostic. On the nonlinear-CMI route the source is $R_C^{\mathrm{src}} = R_C^{\mathrm{info}}$ only after a named CMI-to-modular-source matching theorem or explicit continuation hypothesis with residual ledger.

Target 4.1 (CMI-to-modular-source matching)**.** *The nonlinear-CMI route must prove or declare a versioned hypothesis that, for each small ball $B_\ell(x,u)$,*

$$R_{x,u,\ell}^{\mathrm{info}} = \frac{2\pi}{\hbar c} \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, T_{ab}^A u^a u^b \,\mathrm{d}V + \varepsilon_{\mathrm{match}},$$

with refinement-stable matching residual. Strong subadditivity alone does not prove this statement. With a subtracted background value, positivity requires a separate packet-parent or branch-restriction certificate.

Theorem 4.2 (Conditional local BW source normalization)**.** *Assume the scale hierarchy above, a declared source route, the matching/local modular representation*

$$R^{\mathrm{src}}_{x,u,\ell} = \frac{2\pi}{\hbar c} \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell} T^A_{ab} u^a u^b \, \mathrm{d}V + \varepsilon_{\mathrm{match}},$$

and smooth $\varepsilon_A(x,u) = T^A_{ab}(x) u^a u^b$ *on the ball. Then*

$$R^{\mathrm{src}}_{x,u,\ell} = \frac{8\pi^2 \ell^4}{15\hbar c} \left[\varepsilon_A(x,u) + \frac{\ell^2}{14} D^2_\perp \varepsilon_A(x,u) + O\left(\frac{\ell^4}{L^4_{\mathrm{gradient}}} + \frac{\ell^2}{L^2_{\mathrm{curvature}}} \right) \right] + \varepsilon_{\mathrm{match}}.$$

Hence, on the controlled local branch,

$$\varepsilon_A(x,u) = \frac{15\hbar c}{8\pi^2} \lim \frac{R^{\mathrm{src}}_{x,u,\ell}}{\ell^4}$$

with the stated finite-ball, curvature, and matching corrections.

Proof. The BW ball weight is $w_\ell(r) = (\ell^2 - r^2)/(2\ell)$. In flat Riemann-normal coordinates

$$\int_{B_\ell} w_\ell(r) \, \mathrm{d}^3 x = 4\pi \int_0^\ell r^2 \frac{\ell^2 - r^2}{2\ell} \, \mathrm{d}r = \frac{4\pi \ell^4}{15}.$$

Multiplying by $2\pi/(\hbar c)$ gives $8\pi^2\ell^4/(15\hbar c)$. The next term follows from angular averaging $\varepsilon_A(y) = \varepsilon_A(x) + r^2 D^2_\perp \varepsilon_A(x)/6 + \cdots$ and $\int_{B_\ell} w_\ell(r) r^2 \mathrm{d}^3 x = 4\pi\ell^6/35$, whose ratio to the constant term gives $\ell^2/14$. Curvature corrections are the usual Riemann-normal-coordinate corrections to the volume form and kernel. □

4.3 Localization, Covers, and Tomography

Let $E_{B,\mathrm{phys}}$ be the conditional expectation onto the quotient-visible physical collar algebra. A finite source receipt must bound

$$|R^{\mathrm{src}}_C - \langle E_{B,\mathrm{phys}}(\Delta K) \rangle| \le \varepsilon_{\mathrm{loc}}$$

with hidden-carrier, port-relabeling, repair-schedule, and refinement invariance. Cover independence is a separate requirement: admissible collar covers must converge to the same weak source measure, or a Möbius/inclusion-exclusion valuation must prove that overlapping collars do not double count the same information defect.

A scalar reading for one observer gives only $T^{ab}_A u_a u_b$. To reconstruct a symmetric tensor, the source must be read on enough timelike directions. For $e_x(u) = \lim(15\hbar c/8\pi^2) R^{\mathrm{src}}_{x,u,\ell}/\ell^4$, define the homogeneous extension

$$Q_x(v) = [-g(v,v)] e_x\left(v/\sqrt{-g(v,v)} \right).$$

If Q_x extends quadratically and satisfies the parallelogram identity, there is a unique symmetric T^A_{ab} recovered by polarization. The finite simulator version needs at least ten linearly independent timelike probes and held-out quadraticity tests; a single $u^\mu = (1,0,0,0)$ row is not a stress theorem.

5 The Dark Matter Challenge

In Newtonian gravity and in the weak-field limit of general relativity, a test particle in a circular orbit satisfies

$$\frac{v^2(r)}{r} = g(r) = \frac{GM(< r)}{r^2}. \tag{1}$$

If almost all gravitating matter is visible baryonic matter, then outside the bright disk the enclosed mass should stop growing quickly and the rotation speed should fall as $v(r) \propto r^{-1/2}$. Disk galaxies do not behave that way. Their outer rotation curves often remain flat, so the inferred dynamical mass grows approximately linearly with radius:

$$M_{\rm dyn}(< r) = \frac{v^2 r}{G}. \tag{2}$$

The galaxy problem is only one part of the evidence. Clusters need more gravity than the observed gas and galaxies provide. Weak lensing maps in merging systems such as the Bullet Cluster show gravitational potential peaks separated from the X-ray gas [16]. The CMB and large-scale structure prefer a gravitating component that is mostly non-baryonic, close to pressureless, and present before galaxies settle [5]. Planck's base ΛCDM parameter set gives $\Omega_c h^2 = 0.120 \pm 0.001$ and $\Omega_b h^2 = 0.0224 \pm 0.0001$, so most matter is not baryonic in that model [5].

The challenge includes more than "flat rotation curves". A good theory has to explain:

- galaxy rotation curves and the baryonic Tully-Fisher relation;
- the tight radial acceleration relation, abbreviated RAR;
- galaxy and cluster lensing;
- cluster offsets in mergers;
- CMB acoustic peaks, BAO, matter power, weak lensing, and S_8.

6 Standard Routes

6.1 Cold Dark Matter

The standard route adds a new non-baryonic matter component. In its simplest form it is cold, effectively collisionless, and pressureless on cosmological scales. ΛCDM uses the same component for CMB peaks, large-scale structure, clusters, lensing, and galaxy halos.

The particle identity is absent. Direct detection has produced no confirmed dark matter particle. Galaxy data show a tight connection between baryonic acceleration and observed acceleration, much tighter than a naive halo-plus-disk picture suggests. Halo fitting also introduces galaxy-level nuisance freedom, while the RAR looks like a law.

6.2 MOND

MOND should be treated as a clue, not as a failure. Its basic phenomenology starts from a critical acceleration a_0, and in the deep low-acceleration regime it gives

$$g \simeq \sqrt{a_0 g_b}. \tag{3}$$

For a point baryonic mass this implies

$$v^4 = GM_b a_0, \tag{4}$$

which is exactly the baryonic Tully-Fisher scaling [14]. This is MOND's great success: it goes straight at the galaxy regularity.

The pressure points start when the same static acceleration rule is asked to do more jobs than a galaxy equilibrium law can naturally do:

- *Relativistic completion and lensing.* A nonrelativistic force law does not by itself say which metric photons follow, or how the modified field contributes to the stress side of Einstein's equations. Relativistic completions such as TeVeS exist [15], but the lensing sector then depends on extra fields and closure assumptions.

- *Clusters.* MOND reduces the cluster mass discrepancy but does not remove the need for additional gravitating stress in clusters. Merging systems are especially sharp: if the anomaly is an instantaneous function of the local baryonic acceleration, the lensing peak wants to remain too tightly tied to the visible baryonic source.

- *The early universe.* The CMB acoustic peaks, BAO, matter growth, CMB lensing, and weak-lensing amplitudes need a homogeneous and perturbative gravitating component before settled galaxies exist. A static RAR law does not by itself define that background abundance or its linear response.

- *Non-spherical and time-dependent systems.* The algebraic relation $g \simeq \sqrt{a_0 g_b}$ is clean in high-symmetry systems. In real three-dimensional fields, curl terms, environmental dependence, transport, and relaxation can matter.

OPH does not treat the static RAR formula as the master law. It approximates MOND in the regime where the OPH stress sector has relaxed onto the old, settled, low-acceleration galaxy branch. It need not approximate MOND in mergers, clusters, or the early universe, because those systems are not static one-dimensional galaxy equilibria. The underlying object is an effective stress sector carried by finite overlap repair. In old relaxed disks it projects to a baryon-linked acceleration law. In lensing it enters the metric-potential bookkeeping. In cluster mergers it is transported and relaxes over a finite repair time. In FLRW and perturbation theory it has to be specified as a homogeneous charge, load current, and response kernel.

This is the sense in which the MOND pressure points dissolve in OPH. OPH treats those failures as wrong-regime extrapolations, without making the same algebraic law more complicated. The static galaxy formula is allowed to be a very good equilibrium approximation, while lensing, clusters, and cosmology are handled by the corresponding OPH stress, transport, and perturbation equations. Those equations must pass their own falsifiers. The claim is structural separation, with no immunity from data.

7 The OPH Route

Observer-Patch Holography starts from finite observer patches and the demand that neighboring patches agree on overlap data. The relevant OPH inputs for the dark branch are:

- finite overlap collars and fixed-cutoff collar algebras;

- edge and cut registers on those collars;

- read, compare, repair, commit, and refresh operations;
- a finite carried modular remainder on the collar;
- the recovered Einstein branch with a screen-capacity de Sitter scale.

The dark-sector proposal is:

> The dark source is a transported modular/collar information-defect remainder. It gravitates because it contributes to the effective stress bookkeeping. It is electromagnetically dark because it is a record-repair remainder, not an ordinary luminous matter species.

No new particle species is postulated. The effective dark stress is the gravitating part of finite-observer information repair.

$$G_{ab} + \Lambda g_{ab} = 8\pi G \left(T_{ab}^{\mathrm{SM}} + T_{ab}^{\mathcal{I}} + T_{ab}^{\mathrm{repair}} \right). \tag{5}$$

The symbol A in formulas below denotes this anomaly or information-defect component. It is an effective stress sector, with no literal particle species claim.

This differs from particle dark matter and MOND in a useful way. It is stress-first like dark matter, so lensing and cosmology have a natural slot. It is baryon-linked like MOND only on the settled galaxy branch, because that equilibrium response is activated by collar/cut repair opportunities sourced by baryonic acceleration. The important contrast with MOND is not the square-root galaxy law; OPH intentionally reproduces that approximation where the equilibrium assumptions apply. The contrast is where the law lives. In MOND, the galaxy formula is often treated as the modification itself. In OPH it is the equilibrium projection of a transported stress component. That is why lensing, clusters, and cosmology can have their own equations without giving up the galaxy regularity.

8 Static Galaxy Derivation

8.1 Source From The Carried Collar Remainder

On the support-visible BW branch the primary theorem is the cap automorphism identity. In the special type-I or effective local representation where the geometric generator may be written inside the cap algebra, the OPH modular decomposition has the local form

$$K_C = 2\pi B_C + K_C^{(\mathrm{anom})} + \mathrm{const.} \tag{6}$$

At finite regulator stage, the nonadditive modular part is carried on an overlap collar. The dark branch may use the raw finite diagnostic

$$R_C^{\mathrm{info}} = I_\omega(A : D|B) \geq 0,$$

but the physical density formula uses the routed source variable R_C^{src}. On the fixed-reference route this is the anomalous modular-energy variation. On the nonlinear-CMI route it is raw CMI only after the explicit CMI-to-modular-source matching theorem or hypothesis and its localization, cover, and tomography receipts pass. With those receipts, the effective rest-energy density is written

$$\rho_A c^2 = \frac{15\hbar c}{8\pi^2 \ell^4} R_C^{\mathrm{src}}, \qquad \rho_A = \frac{15\hbar}{8\pi^2 c\, \ell^4} R_C^{\mathrm{src}}. \tag{7}$$

Here ℓ is the proper geodesic radius of the local four-dimensional small ball. It is not a cap angle, a screen spacing, a collar thickness, a coordinate radius, or a fitted galaxy scale. If the simulator records CMI in bits, the value must be multiplied by $\ln 2$ before using (7). The weak-field Poisson equation becomes

$$\nabla^2\Phi = 4\pi G(\rho_b + \rho_A). \tag{8}$$

Conditional theorem 1, attractive sign. If the routed source R_C^{src} is nonnegative on the declared branch, then the anomaly is attractive in the Newtonian weak-field equation. For the nonlinear-CMI route this nonnegativity is inherited from raw CMI only when no background subtraction has been introduced; otherwise it must be certified by the packet parent or branch restriction.

Proof. The premise and (7) give $\rho_A \geq 0$. Equation (8) then shows that the anomaly enters with the same attractive sign as baryonic mass. □

Finite-source receipt boundary. The exact CMI modular identity, first-variation obstruction, declared route, matching theorem or hypothesis, collar localization, BW normalization, ℓ^4 plateau, cover independence, and timelike tomography receipts are the source-side content of Issue #319. Without them, R_C^{info} is a recoverability/readback diagnostic and not a physical density source. After them, a finite positive packet parent must supply T_A^{ab}, its Standard-Model-neutral charge condition, and the conserved anomaly plus recipient stress closure.

8.2 The Acceleration Variable

Define the baryonic acceleration ratio

$$x := \frac{g_b}{a_{0,\mathrm{OPH}}}. \tag{9}$$

For a point baryonic source,

$$g_b(r) = \frac{GM_b}{r^2}, \qquad r_M := \sqrt{\frac{GM_b}{a_{0,\mathrm{OPH}}}}, \tag{10}$$

so

$$x = \left(\frac{r_M}{r}\right)^2, \qquad \sqrt{x} = \frac{r_M}{r}. \tag{11}$$

The OPH support assumption is that settled-galaxy repair opportunities live on codimension-one collar/cut support. A screen-area count would scale as x. A cut-count scales as $\sqrt{x}$, as in (11).

8.3 Poisson Activation

Let the mean number of independent local repair opportunities be

$$\mu(x) = \lambda_{\mathrm{collar}}\sqrt{x}. \tag{12}$$

The Poisson law does not follow from fixing this mean alone. Fixed mean alone would select a geometric distribution on nonnegative integers. The OPH claim needs the stronger collar-count premise: independent increments on disjoint collar intervals, rare local repair events, and refinement stability. Partition an interval I with mean $\mu(I)$ into m equal subintervals. If each subinterval has one activation opportunity with probability $\mu(I)/m + O(m^{-2})$, and the probability of two or more events in one subinterval is $O(m^{-2})$, then for every fixed n,

$$\Pr[N(I) = n] = \lim_{m\to\infty} \binom{m}{n} \left(\frac{\mu(I)}{m}\right)^n \left(1 - \frac{\mu(I)}{m}\right)^{m-n} = e^{-\mu(I)} \frac{\mu(I)^n}{n!}. \tag{13}$$

In particular, the zero-count probability is

$$\Pr[N(I) = 0] = \lim_{m\to\infty} \left(1 - \frac{\mu(I)}{m}\right)^m = e^{-\mu(I)}. \tag{14}$$

The probability that at least one repair opportunity is active is

$$p(x) = 1 - \Pr[N = 0] = 1 - \exp[-\lambda_{\text{collar}}\sqrt{x}]. \tag{15}$$

The activation count is only the counting part of the static law. The source closure says that active scalar repair opportunities recover baryonic flux in the settled one-dimensional branch:

$$g_b = p(x) g_{\text{obs}}. \tag{16}$$

Thus

$$g_{\text{obs}} = \nu_{\text{OPH}}(x) g_b, \qquad \nu_{\text{OPH}}(x) = \frac{1}{1 - \exp[-\lambda_{\text{collar}}\sqrt{x}]}. \tag{17}$$

At $\lambda_{\text{collar}} = 1$ this is the McGaugh–Lelli–Schombert empirical interpolating function fitted to SPARC [8]; SPARC comparisons of (17) are therefore non-held-out. The universal extrapolation of this static law to a Solar source in the Galactic external field is excluded by Cassini at 19.22σ (18.01σ on the unit branch; see the Cassini Applicability Stress subsection). The surviving claim is the declared old-settled-galaxy scope, pending a source-derived applicability theorem.

Conditional theorem 2, activation law. Assume codimension-one collar/cut support, independent increments on disjoint collar intervals, rare local repair events, refinement stability, and the settled scalar flux-recovery closure (16). Then the settled one-dimensional galaxy response is (17).

Proof. Codimension-one support gives a mean count proportional to $\sqrt{x}$. The scalar coefficient is λ_{collar}. Independent increments and rare-event refinement give the Poisson law (13). The active probability is the complement of the zero-count event, which is (15). Solving $g_b = p(x) g_{\text{obs}}$ in one-dimensional symmetry gives (17). □

8.4 Conservative Potential Equation

For general disks and three-dimensional baryonic profiles, the algebraic vector rule

$$\mathbf{g}_{\text{alg}} = \nu_{\text{OPH}}(|\mathbf{g}_b|/a_{0,\text{OPH}}) \mathbf{g}_b \tag{18}$$

need not be curl-free because

$$\nabla \times \mathbf{g}_{\text{alg}} = \nabla \nu_{\text{OPH}} \times \mathbf{g}_b. \tag{19}$$

A gravitational acceleration must be derived from a potential. The primary static OPH equation is therefore

$$\nabla^2 \Phi = \nabla \cdot \left[\nu_{\text{OPH}}\left(\frac{|\nabla \Phi_b|}{a_{0,\text{OPH}}}\right) \nabla \Phi_b\right], \qquad \mathbf{g}_{\text{obs}} = -\nabla \Phi, \tag{20}$$

with

$$\nabla^2 \Phi_b = 4\pi G \rho_b. \tag{21}$$

In one-dimensional symmetry, (20) integrates to

$$g_{\rm obs} = \nu_{\rm OPH}(g_b/a_{0,\rm OPH}) g_b + C. \tag{22}$$

The boundary condition $g_{\rm obs} \to 0$ when the enclosed baryonic flux vanishes sets $C = 0$, so (17) is exact for spherical, planar, or one-dimensional symmetry. The SPARC tables in this note use the algebraic diagnostic; a full disk-potential solver is required for publication-grade rotation and lensing claims. Equation (20) holds only on the declared old-settled-galaxy branch: its universal extension is excluded by the Cassini quadrupole bound at 19.22σ, and no source-derived applicability theorem yet selects the branch.

8.5 Newtonian And Deep-IR Limits

For $x \gg 1$, $p(x) \to 1$, hence

$$g_{\rm obs} \to g_b. \tag{23}$$

For $x \ll 1$,

$$p(x) = \lambda_{\rm collar}\sqrt{x} + O(x), \tag{24}$$

so

$$g_{\rm obs} \simeq \frac{g_b}{\lambda_{\rm collar}\sqrt{g_b/a_{0,\rm OPH}}} = \sqrt{\frac{a_{0,\rm OPH}}{\lambda_{\rm collar}^2} g_b} = \sqrt{a_{0,\rm eff}\, g_b}, \tag{25}$$

where

$$a_{0,\rm eff} = \frac{a_{0,\rm OPH}}{\lambda_{\rm collar}^2}. \tag{26}$$

For a point source,

$$v^4 = G M_b a_{0,\rm eff}. \tag{27}$$

Theorem 3, BTFR. The OPH activation law implies the baryonic Tully-Fisher scaling in the deep-IR limit.

Proof. For a circular orbit, $v^2/r = g_{\rm obs}$. In the deep-IR limit, $g_{\rm obs} = \sqrt{a_{0,\rm eff} G M_b / r^2}$. Multiplying by r gives $v^2 = \sqrt{G M_b a_{0,\rm eff}}$, hence (27). □

8.6 Effective Dark Density

The anomaly can be represented as an equilibrium effective density:

$$\mathbf{g}_A = (\nu_{\rm OPH} - 1)\mathbf{g}_b, \qquad \rho_{A,\rm eq} = -\frac{1}{4\pi G}\nabla \cdot [(\nu_{\rm OPH} - 1)\mathbf{g}_b]. \tag{28}$$

For a spherical system,

$$M_{\rm dyn}(r) = \nu_{\rm OPH}(x) M_b(r), \qquad M_A(r) = [\nu_{\rm OPH}(x) - 1] M_b(r). \tag{29}$$

In the deep-IR point-source limit, $M_{\rm dyn}(r) \propto r$, which is the effective mass profile behind flat rotation curves.

For a point baryonic source, the exact equilibrium expression is

$$M_A(r) = \frac{M_b}{\exp(\lambda_{\text{collar}} r_M/r) - 1}, \tag{30}$$

and

$$\rho_A(r) = \frac{M_b \lambda_{\text{collar}} r_M \exp(\lambda_{\text{collar}} r_M/r)}{4\pi r^4 [\exp(\lambda_{\text{collar}} r_M/r) - 1]^2}. \tag{31}$$

For an extended spherical baryonic profile, positivity of the settled effective density is the monotonicity condition

$$\frac{\mathrm{d}}{\mathrm{d}r}\left([\nu_{\text{OPH}}(x(r)) - 1] M_b(r)\right) \geq 0. \tag{32}$$

Profiles that violate this condition need finite pressure, shear, or transport terms instead of the settled dust equilibrium alone.

RAR/source matching target. The local modular-source normalization and the settled-galaxy RAR activation law are separate claims. The RAR density

$$\rho_A^{\text{RAR}} = \frac{1}{4\pi G} \nabla \cdot [(\nu_{\text{OPH}} - 1) \nabla U_b]$$

is matched to the collar source only after a branch theorem proves

$$\frac{15\hbar c}{8\pi^2 \ell^4} \bar{R}_\ell[\rho_b] = c^2 \rho_A^{\text{RAR}}[\rho_b]$$

with no refit. Until that receipt passes, the RAR law is a settled-equilibrium continuation law, while the BW normalization is the local source theorem.

9 The Finite-Collar Coefficient

9.1 Unit Branch

The default branch is

$$\lambda_{\text{collar}} = 1, \qquad a_{0,\text{eff}} = a_{0,\text{OPH}}. \tag{33}$$

This branch sets the finite-collar opportunity density to one.

9.2 $\mathbb{Z}_6$ Reserve Branch

The OPH gauge structure contains the realized quotient

$$G_{\text{phys}} = \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}. \tag{34}$$

The protected shared-edge reserve has mean

$$\epsilon_{\mathbb{Z}_6} = \frac{\bar{\ell}_{\text{shared}}}{|\mathbb{Z}_6|} = \frac{P/4}{6} = \frac{P}{24}. \tag{35}$$

Using

$$P = 1.630968209403959, \tag{36}$$

gives

$$\epsilon_{\mathbb{Z}_6} = 0.06795700872516496. \tag{37}$$

Trace convention. The exponent mean is the normalized quotient expectation:

$$\tau_q(Z_{6,r,C}) = \epsilon_{\mathbb{Z}_6} = \frac{P}{24}.$$

The reciprocal reserve-depth trace is

$$\mathrm{Tr}_q^{\#}(Z_{6,r,C}) = \frac{24}{P}.$$

The Poisson factor is

$$e^{-\tau_q(Z_{6,r,C})} = e^{-P/24},$$

not $e^{-\mathrm{Tr}_q^{\#}(Z_{6,r,C})}$.

Conditional bridge 4, quotient-edge co-registration. Galaxy scalar repair opportunities are quotient-edge cut-register events on the same edge-center resolution that carries the protected $\mathbb{Z}_6$ reserve. The $\mathbb{Z}_6$ reserve therefore co-registers with the scalar activation count.

Conditional argument. The scalar source is a quotient-local recoverability defect, so pure $\mathbb{Z}_6$ center motion changes no realized matter or Higgs state and is inactive for scalar sourcing. If the scalar count and the protected reserve are forced onto the same center-projector resolution, the reserve removes active scalar opportunities from that same layer. □

Imported scalar-slot closure. This bridge is discharged, on the quotient-edge scalar branch, by the quotient-edge collar theorem stack of the screen-microphysics paper [3]. That paper defines

$$\mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}$$

on the edge-center scalar-completeness branch and proves the imported statements labeled

`thm:scalar-activation-edge-center`, `thm:scalar-z6-same-register-commute`, `thm:scalar-channel-ex`

Thus

$$\Pi^{\mathrm{scal}}_{r,C} \in \mathcal{E}_{r,C}, \qquad Z_{6,r,C} \in \mathcal{E}_{r,C}, \qquad [\Pi^{\mathrm{scal}}_{r,C}, Z_{6,r,C}] = 0,$$

and the scalar activation channel has no independent scalar bulk, screen-area, or noncentral collar carrier unless the scalar-completeness or scalar-activation-channel completeness branch fails. For a nondegenerate OPH-coherent material subfederation, the imported source readout is

$$S^{\mathrm{can}}_{\mathrm{coh}} = \mathbf{1}_{\mathrm{self-read}}\, \mathsf{R}_U\, \mathsf{P}_U\, \mathsf{C}_U > 0$$

and coherent-matter same-channel forcing places this source in $\mathcal{E}_{r,C}$. This is a scalar-channel identity theorem. It supplies no finite covariant dark-stress parent, no abundance prediction, and no Boltzmann-likelihood closure. The scalar source route is

$$R^{\mathrm{src}}_C \longmapsto \mathcal{N}_{\nu,r,C} \longmapsto \nu_{\mathrm{OPH}}.$$

The exact $\mathbb{Z}_6$ coefficient additionally requires the imported gate `UNIFORM_PRODUCT_THICKENING_EXACT`; co-registration alone is not the same as scalar-reserve unbiasedness.

Finite channel bridge. The same-channel statement can be represented by one finite channel object: one indexed family E carries both the record panel $(p_e(q), a_e, q \mapsto q \oplus e)$ used by the scalar opportunity count and the collar panel (w_e, ϵ_e) used by the reserve survival factor. Then the record slots and collar slices are definitionally the same finite family, and the composite law reads

$$\lambda_{\mathrm{collar}}\,(\mathcal{L}^{\mathrm{coh}}\mathcal{N}_\nu)(q) = e^{-P/24}\, S^{\mathrm{can}}_{\mathrm{coh}}(q)\, A(q)$$

on the uniform product-thickening branch. This bridge removes the bookkeeping ambiguity in the phrase "the same counter." It does not close the physical residues: nature must instantiate the one-channel structure, and the numerical record-ΔS to gravity-ΔS calibration remains a separate G9 obligation.

Conditional bridge 5, local Poisson reserve survival. If the reserve occupancy in a co-registered local scalar slot at transverse collar coordinate y is Poisson with mean $\epsilon_{\mathbb{Z}_6}(y)$, then the local scalar survival factor is

$$\lambda_{\mathrm{slot}}(y) = \Pr[N_{\mathbb{Z}_6}(y) = 0] = \exp[-\epsilon_{\mathbb{Z}_6}(y)]. \tag{38}$$

Proof. A Poisson variable with mean $\epsilon_{\mathbb{Z}_6}$ has zero-count probability $\exp(-\epsilon_{\mathbb{Z}_6})$. A scalar opportunity survives only when its co-registered reserve count is zero. This proves the local survival factor. □

9.3 Finite Thickness

Let y be the transverse collar coordinate and $w(y)$ the normalized activation weight:

$$\int \mathrm{d}y\, w(y) = 1. \tag{39}$$

The finite-thickness branch gives

$$\lambda_{\mathrm{collar}} = \int \mathrm{d}y\, w(y) \exp[-\epsilon_{\mathbb{Z}_6}(y)]. \tag{40}$$

Scalar-conditioned reserve mean. The unconditioned reserve mean

$$\tau_q(Z_{6,r,C}) = P/24$$

does not automatically imply

$$\frac{\tau_q(\Pi^{\mathrm{scal}}_{r,C} Z_{6,r,C})}{\tau_q(\Pi^{\mathrm{scal}}_{r,C})} = P/24.$$

The finite-thickness branch uses the scalar-weighted reserve profile

$$\epsilon_{\mathbb{Z}_6}(y) = \frac{\tau(\Pi^{\mathrm{scal}} Z_6 p_y)}{\tau(\Pi^{\mathrm{scal}} p_y)}.$$

Thus the mean condition entering Jensen is

$$\int \mathrm{d}y\, w(y)\epsilon_{\mathbb{Z}_6}(y) = P/24.$$

This is a scalar-weighted mean receipt. It follows on the uniform product-thickening branch, but not from co-registration alone.

If

$$\int \mathrm{d}y\, w(y)\epsilon_{\mathbb{Z}_6}(y) = P/24, \tag{41}$$

then Jensen's inequality gives

$$\lambda_{\mathrm{collar}} \geq \exp(-P/24). \tag{42}$$

Equality holds when $\epsilon_{\mathbb{Z}_6}(y)$ is constant on the weighted collar.

Conditional bridge 6, exact finite-thickness coefficient. The exact finite-thickness coefficient requires the uniform product-thickening branch of the screen-microphysics paper:

$$\mathcal{A}_{r,C}^{\mathrm{thick}} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y),$$

$$\tau_{q,r,C}^{\mathrm{thick}} = \tau_{q,r,C}^{\mathcal{E}} \otimes \tau_r^{Y},$$

$$Z_{6,r,C}^{\mathrm{thick}} = Z_{6,r,C} \otimes 1_Y,$$

and, essentially,

$$\frac{\tau_{q,r,C}^{\mathcal{E}}\Big(\Pi_{r,C}^{\mathrm{scal}}(y) Z_{6,r,C}\Big)}{\tau_{q,r,C}^{\mathcal{E}}\Big(\Pi_{r,C}^{\mathrm{scal}}(y)\Big)} = \frac{P}{24}$$

for every scalar-active transverse slice y. Product trace alone is not enough. The scalar-active sampler must be reserve-unbiased.

Conditional argument. With local Poisson reserve survival and the uniform product-thickening branch, the imported theorem labeled `thm:exact-uniform-product-thickening-coefficient` gives

$$\lambda_{\mathrm{collar}} = \exp(-P/24) = 0.9343006394893864\ldots. \tag{43}$$

Without the uniform branch, the theorem-grade coefficient is

$$\lambda_{\mathrm{collar}} = \int \mathrm{d}y\, w(y) e^{-\epsilon_{\mathbb{Z}_6}(y)}.$$

If the scalar-weighted protected reserve mean is $P/24$, Jensen gives only

$$e^{-P/24} \leq \lambda_{\mathrm{collar}} \leq 1.$$

On that exact branch,

$$\lambda_{\mathrm{collar}} = \exp(-P/24) = 0.934300639489386\ldots \tag{44}$$

and by (26)

$$a_{0,\mathrm{eff}} = 1.179018696 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{45}$$

Failure consequence. Without finite-thickness uniformity, the coefficient is the integral (40). Exact $\exp(-P/24)$ is not available as a theorem-grade number.

10 Lensing And Relativistic Stress

A density source by itself fixes the Newtonian Poisson equation. Lensing also depends on the spatial stress. The OPH no-slip claim uses the transported record-number stress and its spatial moments.

The conditional OPH stress closure is a kinetic record measure. At each coarse-grained event, the finite collar packet emits a positive measure $\mathrm{d}\mathcal{E}_x(v)$ on the future unit timelike shell. Define

$$J^a_{\mathcal{I}}(x) = \int v^a \, \mathrm{d}\mathcal{E}_x(v), \qquad T^{ab}_{\mathcal{I}}(x) = \int v^a v^b \, \mathrm{d}\mathcal{E}_x(v). \tag{46}$$

This stress is symmetric and future-positive. Relative to an observer u^a, it decomposes as

$$T^{ab}_{\mathcal{I}} = \epsilon_{\mathcal{I}} u^a u^b + P_{\mathcal{I}} h^{ab} + 2u^{(a} q^{b)}_{\mathcal{I}} + \pi^{ab}_{\mathcal{I}}. \tag{47}$$

The settled branch should not be modelled as an exactly static extended monokinetic dust halo. Separately conserved dust follows geodesics and cannot generically remain static in a nonconstant gravitational potential. The settled no-slip branch is instead a cold-isotropic kinetic branch

$$f_A(x, \mathbf{p}) = F(x, |\mathbf{p}|)$$

in the local rest frame, or a stationary $F(E)$ branch when an energy integral exists. Angular symmetry gives $q^a_{\mathcal{I}} = 0$ and $\pi^{ab}_{\mathcal{I}} = 0$, while a small pressure or velocity dispersion may support the halo. The cold condition is

$$0 \le \frac{P_{\mathcal{I}}}{\epsilon_{\mathcal{I}}} \le \epsilon_{\mathrm{cold}} \ll 1,$$

with a Jeans or stationary-Vlasov residual such as

$$|\nabla_i P_{\mathcal{I}} + \rho_{\mathcal{I}} \nabla_i U| \le \tau_{\mathrm{Jeans}}$$

in the Newtonian regime. In Newtonian gauge,

$$\mathrm{d}s^2 = a^2[-(1 + 2\Psi)\mathrm{d}\eta^2 + (1 - 2\Phi)\mathrm{d}x^2]. \tag{48}$$

The off-diagonal spatial Einstein equation gives

$$k^2(\Phi - \Psi) = 12\pi G a^2 \sum_i (\rho_i + P_i)\sigma_i. \tag{49}$$

The $\mathcal{I}$-sector intrinsic contribution to the right-hand side vanishes whenever the transported packet distribution is rotationally invariant in the local rest frame, so $q^a_{\mathcal{I}} = 0$ and $\pi^{ab}_{\mathcal{I}} = 0$. A velocity-spread or imperfectly isotropic branch must instead emit

$$\epsilon_\pi = \frac{|\pi_A|}{\epsilon_A + P_A}$$

and an $O(\epsilon_\pi)$ slip bound. If baryonic, neutrino, radiation, and repair-recipient anisotropic stresses are absent or handled separately, if there is one metric source, and if boundary conditions do not add an unmodelled transverse exchange, then

$$\Phi = \Psi. \tag{50}$$

The lensing potential is therefore the same potential that controls dynamics:

$$\Phi_{\mathrm{lens}} = \frac{\Phi + \Psi}{2} = \Phi. \tag{51}$$

In weak-field physical units, with $U = c^2\Phi = c^2\Psi$ on this no-intrinsic slip branch,

$$\nabla^2 U = 4\pi G(\rho_b + \rho_A) + \mathcal{E}_{\rm PN} + \mathcal{E}_P + \mathcal{E}_{\rm match} + \mathcal{E}_{\rm finite,ball},$$

and geometric-optics lensing uses

$$\boldsymbol{\alpha} = \frac{2}{c^2}\int \nabla_\perp U \, \mathrm{d}z.$$

A simulator that fits separate dynamical and lensing potentials has not passed the common-metric receipt.

This is where OPH behaves differently from a force-only MOND law. The correction is attached to stress bookkeeping, so the same potential controls slow matter and lensing on settled no-slip galaxy states. Cluster mergers and cosmological perturbations use the full decomposition (47); a velocity-spread branch can carry pressure, heat flux, or anisotropic stress.

11 Clusters And Transport

The static galaxy law is an equilibrium law. Merging clusters need a transported information-defect state. In the cold transported branch, the kinetic stress (46) reduces to a density and velocity field:

$$\partial_t \rho_{\mathcal{I}} + \nabla\cdot(\rho_{\mathcal{I}}\mathbf{v}_{\mathcal{I}}) = -\frac{\rho_{\mathcal{I}} - \rho_{\mathcal{I},\rm eq}[\rho_b]}{\tau_{\rm rec}}, \tag{52}$$

$$\partial_t \mathbf{v}_{\mathcal{I}} + (\mathbf{v}_{\mathcal{I}}\cdot\nabla)\mathbf{v}_{\mathcal{I}} = -\nabla\Phi, \tag{53}$$

$$\nabla^2\Phi = 4\pi G(\rho_b + \rho_{\mathcal{I}}). \tag{54}$$

Velocity-spread branches use the moment equations of (47) instead of (52).

The repair-commit theorem gives the form of the relaxation term. A transition matrix has a dimensionless step eigenvalue. Physical relaxation uses its logarithmic decay together with the physical commit time. Let $r_\star$ be the largest nonstationary step eigenvalue in modulus, $g_{\rm step} = 1 - r_\star$, and $\gamma_{\log} = -\log|r_\star|$. With physical commit time $t_{\rm commit}$,

$$\Gamma_{\rm rec} = \frac{\gamma_{\log}}{t_{\rm commit}}, \qquad \tau_{\rm rec} = \frac{t_{\rm commit}}{\gamma_{\log}}. \tag{55}$$

The small-gap approximation is $\Gamma_{\rm rec} = g_{\rm step}/t_{\rm commit} + O(g_{\rm step}^2)$. The symbols $r_\star$, $g_{\rm step}$, $\gamma_{\log}$, and $\Gamma_{\rm rec}$ denote different artifact types.

Bianchi consistency requires a repair-exchange stress:

$$\nabla_a(T_b^{ab} + T_{\mathcal{I}}^{ab} + T_R^{ab} + \cdots) = 0. \tag{56}$$

In quasi-covariant notation,

$$\nabla_a T_{\mathcal{I}}^{ab} = -\Gamma_{\rm rec}(\rho_{\mathcal{I}} - \rho_{\mathcal{I},\rm eq})u_{\mathcal{I}}^b, \tag{57}$$

and T_R^{ab} carries the opposite exchange flux.

An OPH-side no-fit clock candidate is the Jacobi mismatch rate. Let $E_{ij} = C_{0i0j}$ be the electric Weyl tidal tensor measured in the local free-fall frame, equivalently the trace-free Newtonian Hessian in the weak field. The homogeneous FLRW Ricci focusing term is subtracted, because it does not create a local overlap-visible cluster offset. The scalar rate

$$\Gamma_J = \left(\frac{E_{ij}E^{ij}}{6}\right)^{1/4}, \qquad \tau_J = \Gamma_J^{-1} \tag{58}$$

is invariant under local rotations and has the right units. For a Newtonian point source,

$$E_{ij}E^{ij} = 6\left(\frac{GM}{r^3}\right)^2, \qquad \tau_J = \sqrt{\frac{r^3}{GM}}. \tag{59}$$

The numerical scale is:

Scale	$M/M_\odot$	r in kpc	τ_J in Gyr
Inner spiral disk	6.0×10^{10}	10	0.0608673
Outer spiral disk	6.0×10^{10}	30	0.316276
Dwarf scale	1.0×10^{9}	5	0.166692
Cluster core	1.0×10^{14}	300	0.244986
Massive cluster aperture	1.0×10^{15}	1000	0.471476

For $d_0 = 200\,\mathrm{kpc}$ and $t = 0.2\,\mathrm{Gyr}$, the massive-cluster aperture row gives $d(t) = 130.859\,\mathrm{kpc}$.

The conditional physical-time conversion is

$$\Gamma_{\mathrm{rec}}(x,t) = \Gamma_J(x,t), \qquad t_{\mathrm{commit}}(x,t) = \frac{\gamma_{\log}(x,t)}{\Gamma_J(x,t)}. \tag{60}$$

The static RAR law does not determine this clock. Any positive relaxation rate leaves the same equilibrium density. The Jacobi clock requires the premise that repair timing is set by the overlap-visible tidal mismatch rate of transported information-defect records. The theorem-grade packet clock must derive this premise without using cluster offsets as selectors.

For a one-mode merger offset,

$$d(t) = d_0 \exp(-t/\tau_{\mathrm{rec}}), \qquad \tau_{\mathrm{rec}} = \frac{t}{\ln(d_0/d(t))}. \tag{61}$$

With $d_0 = 200\,\mathrm{kpc}$ and $t = 0.2\,\mathrm{Gyr}$:

Retained offset	Required τ_{rec}
50 kpc	0.144 Gyr
100 kpc	0.288 Gyr
134 kpc	0.500 Gyr
150 kpc	0.696 Gyr
164 kpc	1.000 Gyr

Settled galaxies impose

$$\tau_{\mathrm{rec}} < \frac{T_{\mathrm{set}}}{\ln(1/\epsilon)} \tag{62}$$

if the allowed tracking error after settling time T_{set} is ϵ. For $T_{\mathrm{set}} = 5\,\mathrm{Gyr}$, the bounds are 2.17 Gyr, 1.67 Gyr, and 1.09 Gyr for 10%, 5%, and 1% tracking error.

12 CMB, BAO, Growth, And S_8

The linear cosmology contract has two distinct branches. A conserved homogeneous information-defect branch obeys

$$\rho_A' + 3\mathcal{H}\rho_A = 0, \qquad \rho_A(a) = \rho_{A0}a^{-3}, \tag{63}$$

where primes denote conformal time and $\mathcal{H} = a'/a$. This branch behaves like a cold stress component if its kinetic measure is monokinetic and its homogeneous charge $Q_A = a^3 \rho_A V_{\rm com}$ is transported by an anomaly-current conservation receipt and selected by the source-only anomaly abundance receipt of Section 17.15. A source-localized CMI value without those receipts does not imply the a^{-3} scaling or the homogeneous amplitude.

No-go boundary for scalar parents. A scalar parent that emits only $\rho_A(a)$, $\rho_{A,\rm eq}(a)$, and $B_A(k,a)$ is a source candidate, not a physical Boltzmann closure. It does not determine

$$w_A(a), \quad c^2_{s,A}(k,a), \quad \sigma_A(k,a), \quad Q^\mu_A, \quad \Gamma_{\rm rec}(k,a),$$

and it does not by itself fix gauge-independent perturbations or the recipient stress needed when repair transfers energy-momentum. In the physical source contract B_A is built from the anomaly-frame baryon density

$$n_b^{(A)} = -u_{A\mu} J_b^\mu,$$

not from a gauge-fixed coordinate density. The physical object is a finite covariant collar-packet parent

$$\mathcal{P}_r[X,g] = (C_r, Z_r, A_r, R_r, G_r, \pi_r[X], L_r[X], Q_r, D_r),$$

where Z_r is a finite packet-state set split into anomaly and recipient channels, π_r is the convex equilibrium packet functional, L_r is the repair generator, Q_r records reaction channels, and D_r records any causal auxiliary response. The parent must emit the stress moments and source projections consumed by the Boltzmann solver; otherwise CMB/BAO/growth rows remain diagnostics.

Finite covariant parent object. The theorem-grade parent is not a parent-shaped JSON object. At regulator r it is a quotient-visible finite kinetic package

$$\mathcal{P}_r[X,g] = (\mathcal{C}_r, g_r, e_r, U_r, Z_r, \omega_r, p_r, f_r, \Phi_r, \mathcal{R}_r, \mathcal{G}_r, \pi_r, \mathsf{Read}_r),$$

with finite causal complex $\mathcal{C}_r$, local tetrads e_r, finite parallel transports U_r, packet sectors

$$Z_r = \bigsqcup_s Z_{s,r},$$

positive quadrature weights, occupations $f_{c,z} \geq 0$, and momenta obeying the declared $(-+++)$ mass shell

$$g_{ab} p^a_z p^b_z = -m^2_z, \qquad p^0_z > 0.$$

The quotient $\mathcal{G}_r$ includes hidden carrier coordinates, equivalent packet labels, port relabelings, local-frame presentations, and accepted repair-schedule presentations. The readout

$$\mathsf{Read}_r = (\mathsf{StressRead}_r, \mathsf{ExchangeRead}_r, \mathsf{ScalarSourceMap}_r, \mathsf{ClockRead}_r, \mathsf{RepairTangent}_r)$$

must factor through that quotient.

Exact finite closure receipt. For each sector define

$$J^a_{s,c} = \frac{1}{V_c} \sum_{z \in Z_s} W_{c,z} p^a_z, \qquad T^{ab}_{s,c} = \frac{1}{V_c} \sum_{z \in Z_s} W_{c,z} p^a_z p^b_z.$$

The simulator must reconstruct these moments from packet rows and emit residual ledgers

$$R^a_{s,c} = (D_r T_s)^a - Q^a_{s,c}, \qquad R^a_{\mathrm{total},c} = D_r T^a_{\mathrm{total}},$$

including internal-face cancellation, boundary flux, geometry/redshift work, external work, reaction four-momentum, and conserved charges. Producer booleans, zero-density labels, invented recipient rows, or scalar residuals are not stress closure. Promoting $\rho_A(a)$, $B_A(k,a)$, a certified $\Gamma_{\mathrm{rec}}(k,a)$, or any associated primordial scalar input also requires a source-provenance DAG and a pooled-reducer receipt. A transition spectral number remains $\gamma_{\mathrm{repair\ step}}$ until active-fiber, physical-clock, and common-parent response receipts pass. Contradictory no-data-use metadata fails closed, and nonlinear quantities such as amplitudes, ranks, condition numbers, and isocurvature leakage are computed only after additive sufficient statistics are pooled globally.

For the source-only primordial bridge, this closed branch has the stricter contract

$$\nabla_a J^a_A = 0, \qquad T^{ab}_A = \epsilon_A u^a u^b, \qquad Q^a_A = 0. \tag{64}$$

It may enter the rank-one primordial theorem only when ρ_A is a function of the same source clock as the rest of the total stress. Transport of a supplied Q_A is not an OPH derivation of the homogeneous abundance.

A repair-exchange branch instead relaxes toward a finite-collar equilibrium source:

$$\rho'_A + 3\mathcal{H}\rho_A = -a\Gamma_{\mathrm{rec}}(\rho_A - \rho_{A,\mathrm{eq}}). \tag{65}$$

The factor $a\Gamma_{\mathrm{rec}}$ is the conformal-time rate associated with the physical repair rate. Define

$$q_A = \frac{\rho_{A,\mathrm{eq}}}{\rho_A}, \qquad \delta_{A,\mathrm{eq}}(k,a) = B_A(k,a)\delta_b(k,a). \tag{66}$$

For a pressureless no-slip repair branch with energy exchange parallel to the anomaly four-velocity, the Newtonian-gauge perturbations reduce to

$$\delta'_A = -\theta_A + 3\Phi' - a\Gamma_{\mathrm{rec}} q_A(\delta_A - \delta_{A,\mathrm{eq}}), \tag{67}$$
$$\theta'_A = -\mathcal{H}\theta_A + k^2\Psi. \tag{68}$$

The no-slip statement here concerns the information-defect sector. The total metric slip depends on baryons, radiation, neutrinos, repair stress, and any anisotropic stress in a velocity-spread branch.

For the same primordial bridge, the repair-exchange branch is admissible only with an explicit transfer receipt:

$$\nabla_a T^{ab}_A = Q^b_A, \qquad \Xi^A_a = 0, \qquad h^a{}_b Q^b_A = 0 \tag{69}$$

or with interval bounds on the two defects. The dark-sector continuation in this paper separates transport of a supplied homogeneous load from source selection of that load. Transport alone has label

`PHYSICAL_PARENT_WITH_CONDITIONAL_ABUNDANCE.`

A source-only homogeneous abundance requires

`RHO_A_SOURCE_RECEIPT` $=$ `RHO_A_TRANSPORT_RECEIPT` $\wedge$ `ANOMALY_ABUNDANCE_SOURCE_RECEIPT.`

For primordial certification without that receipt, use

`dark_continuation = OFF.`

A supplied dark abundance may be used only as

```
dark_continuation = CONDITIONAL_SOURCE_STATE,
```

not as a derived OPH primordial prediction.

If the exchange branch is nonzero, the recipient sector is not optional. The finite parent must expose a recipient stress T_R^{ab}, a reaction channel, and equal-and-opposite exchange currents such that

$$\nabla_a(T_A^{ab} + T_R^{ab} + T_{\text{other}}^{ab}) = 0.$$

Without that explicit stress and current closure, a repair sink/source can mimic a scalar damping term but cannot be promoted to an Einstein-source prediction.

The finite scalar-load repair diagnostic gives

$$\gamma_{\text{repair step}} = 0.050717471.$$

It maps a cluster-scale relaxation window to the following commit times:

τ_{rec}	t_{commit}
0.3 Gyr	15.2152 Myr
0.5 Gyr	25.3587 Myr
1.0 Gyr	50.7175 Myr

For the same relaxation window, a simple background timing diagnostic gives:

τ_{rec}	$z = 0$	$z = 10$	$z = 1100$
0.3 Gyr	48.4092	2.36213	0.00205644
0.5 Gyr	29.0455	1.41728	0.00123387
1.0 Gyr	14.5228	0.708639	0.000616933

For a hybrid branch, repair relaxation is tiny during acoustic physics and large in low-Hubble settled environments. A cosmology claim requires a finite covariant source artifact, Boltzmann implementation, frozen likelihood hashes, and likelihood tests against CMB TT/TE/EE, CMB lensing, BAO, weak lensing, redshift-space distortions, and S_8.

13 Abundance And Kernel Targets

The static galaxy law fixes a settled response around resolved baryonic inhomogeneity. A cosmological dark-matter branch needs OPH source outputs:

$$\rho_A(a), \qquad B_A(k,a), \qquad \Gamma_{\text{rec}}(k,a), \qquad w_A,\ c_{s,A}^2,\ \sigma_A,\ Q_A^\mu. \tag{70}$$

The first is the homogeneous anomaly abundance. The second is the linear equilibrium-source kernel in

$$\delta_{A,\text{eq}}(k,a) = B_A(k,a)\delta_b(k,a). \tag{71}$$

The remaining entries close the stress tensor and exchange law.

13.1 Static FLRW Boundary

The nonlinear galaxy law cannot be inserted directly into FLRW perturbation theory. In an exactly homogeneous background there is no preferred baryonic acceleration vector. If one tries to Taylor-expand the static RAR law at $\mathbf{g}_b = 0$, the susceptibility is singular because $\nu_{\mathrm{OPH}}(x) \sim (\lambda_{\mathrm{collar}}\sqrt{x})^{-1}$. The static RAR branch therefore does not select a homogeneous anomaly density or a universal CMB kernel.

This is a useful boundary. It prevents a hidden fit from being smuggled into the galaxy law. The CMB branch has to declare its background anomaly charge and its perturbation kernel from OPH state selection or finite-collar microphysics.

13.2 Capacity-Saturated State Selection

One clean homogeneous benchmark is flat capacity saturation. More generally, the de Sitter screen capacity fixes H_{dS}; the capacity itself is the SL-4 estimate read from measured Λ with the scale bridge (`../STRANGE_LOOP_PRINCIPLES.md`), so H_{dS} is an input-carrying quantity rather than an OPH output. The object that would promote this capacity lane is generator G2 of `../CONSISTENCY_STACK.md`: constructing the readback map F and its contraction certificate. The Friedmann closure at $a = 1$ gives only the degenerate combination

$$\Omega_{A0} + \Omega_{K0} = 1 - \Omega_{\Lambda,\mathrm{OPH}} - \Omega_{b0} - \Omega_{\nu 0} - \Omega_{r0}, \qquad \Omega_{\Lambda,\mathrm{OPH}} = \left(\frac{H_{\mathrm{dS}}}{H_0}\right)^2. \tag{72}$$

The displayed number becomes an Ω_{A0} value only after an independent curvature readout or an explicit flat-branch assumption $\Omega_{K0} = 0$. It cannot be used to prove flatness. The transported homogeneous dust branch is

$$\rho_A(a) = \rho_{A0} a^{-3}. \tag{73}$$

The legacy plumbing run used $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_b h^2 = 0.02237$, $\Omega_r = 9.17 \times 10^{-5}$, and the target-informed weighted-cycle comparison value $\sum m_\nu = 0.0900119296\,\mathrm{eV}$. That neutrino value is now rejected and has no OPH prediction status. The resulting historical residual was:

Quantity	Legacy diagnostic value
H_{dS}	$55.759940256\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
$\Omega_{\Lambda,\mathrm{OPH}}$	0.684423332
Ω_b	0.049243191
Ω_ν	0.002127375
Ω_r	0.000091700
$\Omega_A + \Omega_K$	0.264114401
Ω_m	0.315484968
ρ_{A0}	$2.253649933 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$

The $\Omega_m = 0.315484968$ here is the Friedmann closure sum $\Omega_A + \Omega_K + \Omega_b + \Omega_\nu$; the CAMB tables later in the paper print the solver output 0.315905207 from the same inputs, and the quoted Planck pull uses the latter. This legacy number matches the Planck-like cold-matter budget at the compressed-parameter level only on the explicit flat benchmark. It is invalidated as a current quantitative OPH row because its supplied neutrino input came from the rejected weighted-cycle candidate. A current run must declare an external neutrino scenario and recompute the residual. Even then, the number is a closure residual, not an independent anomaly abundance and not a flatness proof. The static galaxy RAR law does not determine it. The OPH theorem that would make Ω_{A0} standalone is a finite-screen additive-load selector for the conserved homogeneous anomaly charge together with an independent curvature label.

Finite exports must therefore label

$$\Omega_{A,\mathrm{source}} \in \{\textsc{IndependentOPH}, \textsc{External}, \textsc{ResidualDefinition}, \textsc{Unknown}\}$$

and separately label the curvature source. A residual-defined Ω_A cannot pass a Friedmann-curvature independence audit.

The closed transported branch does give a theorem-grade current and charge. Let u_A^a be the branch four-velocity and let ρ_A be the coarse-grained information-defect load density. The load current is

$$J_A^a = \rho_A u_A^a. \tag{74}$$

On the closed branch,

$$\nabla_a J_A^a = 0, \qquad Q_A[\Sigma] = \int_\Sigma J_A^a n_a \,\mathrm{d}\Sigma \tag{75}$$

is independent of the homogeneous slice. In FLRW this gives

$$Q_A = a^3 \rho_A V_{\mathrm{com}}, \qquad \rho_A(a) = \frac{Q_A}{a^3 V_{\mathrm{com}}}. \tag{76}$$

This proves transport of a supplied homogeneous load. It does not fix the amplitude Q_A. The source-only selector is the quotient expectation of the parent-emitted anomaly load on a release hypersurface,

$$\mathcal{E}_{A,r} = F_{A,r}\Big(Q^{\mathrm{rel}}_{A,r}, \mu^{\mathrm{rel}}_{A,r}, P_\star, N_{\mathrm{CRC}}, \mathfrak{R}^{\mathrm{rel}}_{A,r}, \mathcal{P}_r\Big), \qquad \rho_A(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\mathrm{com},r}}. \tag{77}$$

Section 17.15 gives the finite selector and receipt. Without that source selector receipt, the abundance is cosmological state data or the flat capacity-saturated residual (72).

13.3 Environmental Kernel

Around a nonzero local background field $\bar{\mathbf{g}}_b$, write

$$x = \frac{|\bar{\mathbf{g}}_b|}{a_{0,\mathrm{OPH}}}, \qquad \mathbf{n} = \frac{\bar{\mathbf{g}}_b}{|\bar{\mathbf{g}}_b|}. \tag{78}$$

Linearizing the anomaly field $(\nu_{\mathrm{OPH}} - 1)\mathbf{g}_b$ gives the susceptibility

$$A_{ij} = [\nu_{\mathrm{OPH}}(x) - 1]\delta_{ij} + x\nu'_{\mathrm{OPH}}(x) n_i n_j, \tag{79}$$

where

$$\nu'_{\mathrm{OPH}}(x) = -\frac{\lambda_{\mathrm{collar}} \exp[-\lambda_{\mathrm{collar}}\sqrt{x}]}{2\sqrt{x}[1-\exp(-\lambda_{\mathrm{collar}}\sqrt{x})]^2}. \tag{80}$$

For a Fourier mode whose direction has cosine μ against $\mathbf{n}$, the density-response multiplier is

$$K_A^{(\rho)}(x,\mu) = \nu_{\mathrm{OPH}}(x) - 1 + x\nu'_{\mathrm{OPH}}(x)\mu^2. \tag{81}$$

An isotropic environmental average gives

$$\langle K_A^{(\rho)} \rangle = \nu_{\mathrm{OPH}}(x) - 1 + \frac{x}{3}\nu'_{\mathrm{OPH}}(x). \tag{82}$$

The contrast kernel used in perturbation equations is normalized by the equilibrium-background density ratio only after the response is diagonal in the calibrated physical mode basis:

$$B_A(k,a) = \frac{\bar\rho_b(a)}{\bar\rho_{A,\mathrm{eq}}(a)} K_A^{(\rho)}(k,a). \tag{83}$$

Using $\bar\rho_A$ in the denominator would define the different variable $\delta\rho_{A,\mathrm{eq}}/\bar\rho_A$ and would require rederiving the external $q_A = \bar\rho_{A,\mathrm{eq}}/\bar\rho_A$ factor in the perturbation equations. This paper uses the equilibrium-contrast convention globally. At finite regulator the primary object is the response operator

$$(B_{A,r})_{jj'}(a) = \frac{\bar\rho_b(a)}{\bar\rho_{A,\mathrm{eq}}(a)} \left\langle v_{rj}, D\rho_{A,\mathrm{eq}}[\bar\rho_b]\, v_{rj'} \right\rangle, \tag{84}$$

where v_{rj} are the same calibrated comoving scalar modes used by the primordial source and transfer handoff. A scalar table $B_A(k_j,a)$ is promoted only if

$$\|[B_{A,r}, \Delta_{\mathrm{com},r}]\| \le \varepsilon_{\mathrm{comm}}, \qquad \sum_{j'\neq j} |(B_{A,r})_{jj'}|^2 \le \varepsilon_{\mathrm{off}}^2$$

on the resolved band. Otherwise the physical input is the mode-coupling matrix, not a diagonal function of k. In a realistic cosmological calculation, this local expression has to be integrated against the distribution of environmental fields,

$$K_A^{(\rho)}(k,a) = \int \mathrm{d}x\, \mathrm{d}\mu\, \Pi(x,\mu|k,a)[\nu_{\mathrm{OPH}}(x) - 1 + x\nu'_{\mathrm{OPH}}(x)\mu^2]. \tag{85}$$

This average is finite only under an explicit small-field support condition, for example

$$\int_0^{\epsilon} \mathrm{d}x\, \mathrm{d}\mu\, \Pi(x,\mu|k,a)\, x^{-1/2} < \infty \qquad \text{for some } \epsilon > 0. \tag{86}$$

If Π has an exact atom at $x = 0$, the local nonzero-field kernel is not the correct FLRW response. The distribution Π is an OPH finite-collar output or an explicitly declared environmental closure. Exact FLRW has $x = 0$, so (81) does not provide a regular universal CMB density-response kernel by itself.

The implementation contract uses one parent finite-collar functional:

$$\rho_{A,\mathrm{eq}}[X]c^2 = \frac{15}{8\pi^2 \ell(X)^4} \int_{\mathcal{C}_X} \mathrm{d}\mu_C\, R_C^{\mathrm{src}}[\omega_C]. \tag{87}$$

For finite samples s with weights w_s, define

$$R(X) = \frac{\sum_s w_s R_s^{\mathrm{src}}}{\sum_s w_s}, \tag{88}$$

$$K_A^{(\rho)}(k,a) = \frac{1}{\bar\rho_b} \frac{\partial \rho_{A,\mathrm{eq}}}{\partial \delta_b(k,a)}, \tag{89}$$

$$B_A(k,a) = \frac{\bar\rho_b(a)}{\bar\rho_{A,\mathrm{eq}}(a)} K_A^{(\rho)}(k,a). \tag{90}$$

The same evaluator emits the homogeneous equilibrium density, the linear density-response kernel, the settled galaxy source, and the cluster equilibrium source. A theorem-grade prediction requires the OPH collar measure $\mathrm{d}\mu_C$, the finite-screen ensemble, and the small-field support condition. Fitting Π to CMB, weak-lensing, SPARC, or cluster data is excluded.

13.4 Physical Source And Kernel Promotion

The scalar tables used by Boltzmann codes are the last stage of a finite source construction. Let $Q_{r,a}$ be the finite quotient state space at regulator r and background scale factor a. A physical anomaly source is a quotient-visible C^1 functional

$$\mathcal{E}_{A,r} : Q_{r,a} \times \mathcal{A} \longrightarrow \mathbb{R} \tag{91}$$

such that

$$\bar{\rho}_{A,\mathrm{eq},r}(a) = \mathcal{E}_{A,r}(q_{0,r}(a), a), \qquad D\mathcal{E}_{A,r}|_{q_0} \text{ is the anomaly response used in } B_{A,r}. \tag{92}$$

For the fixed-reference modular-energy route one may write, on each frozen reference patch,

$$\mathcal{E}_{A,r}(q, a) = \rho^{(0)}_{A,\mathrm{eq},r}(a) + \mathcal{N}_{r,a}\ \mathrm{Tr}\big[\big(\rho_r(q) - \sigma_{r,a}\big)K^{\mathrm{anom}}_{\sigma,r,a}\big] + \mathcal{R}_{2,r}(q, a), \tag{93}$$

with $\mathcal{R}_{2,r}(q_0, a) = 0$ and $D\mathcal{R}_{2,r}|_{q_0} = 0$. Then

$$D\mathcal{E}_{A,r}|_{q_0}[v] = \mathcal{N}_{r,a}\,\mathrm{Tr}\big[D\rho_r|_{q_0}[v]\, K^{\mathrm{anom}}_{\sigma,r,a}\big]\,.$$

The shift $K^{\mathrm{anom}}_{\sigma,r,a} \mapsto K^{\mathrm{anom}}_{\sigma,r,a} + cI$ leaves this tangent term invariant because $\mathrm{Tr}\, D\rho_r[v] = 0$. The baseline $\rho^{(0)}_{A,\mathrm{eq},r}$ is separate finite source data; a modular variation alone gives a response, not an absolute positive abundance. For a nonlinear-CMI route at an exact Markov reference, the first variation vanishes and the leading object is the Hessian. A nonzero linear response from CMI therefore requires a declared non-Markov background and its residual.

Let finite packet occupations satisfy

$$\mathcal{M}_{r,a} = \{f \in \mathbb{R}^{N_r}_{\geq 0} : A_{r,a} f = b_{r,a}\}, \qquad Q_{r,a} = \mathcal{M}_{r,a}/\mathcal{G}_{r,a}.$$

At an interior background $f_0 > 0$,

$$T_{[f_0]} Q_{r,a} = \ker A_{r,a}/T_{f_0}(\mathcal{G}_{r,a} \cdot f_0). \tag{94}$$

Boundary backgrounds use the corresponding tangent cone. A symmetric $\pm\epsilon$ finite difference is admissible only when both perturbed states stay in the same admissible support. For a retraction $\mathcal{R}_{q_0}(0) = q_0$, $D\mathcal{R}_{q_0}(0) = I$, and a C^3 readout E_r,

$$\frac{E_r(\mathcal{R}_{q_0}(\epsilon v)) - E_r(\mathcal{R}_{q_0}(-\epsilon v))}{2\epsilon} = DE_r[v] + O(\epsilon^2).$$

If the third directional derivative is bounded by M_v, the truncation error is at most $M_v \epsilon^2/6$.

Let $S_r : Q_{r,a} \to U_r$ be the delivered baryon-source map and $E_r : Q_{r,a} \to Y_r$ the anomaly source map. A physical response depending only on the delivered source exists exactly when

$$\ker DS_r|_{q_0} \subseteq \ker DE_r|_{q_0}. \tag{95}$$

Then there is a unique linear operator

$$B_{A,r} : \mathrm{im}\, DS_r|_{q_0} \to Y_r$$

such that $DE_r|_{q_0} = B_{A,r} DS_r|_{q_0}$. This is the lift-independence condition: perturbing a collar width, packet occupation, or port current gives the same kernel only if those microscopic lifts deliver the same physical source tangent.

The delivered source should initially be the vector

$$\mathcal{S}_b = \big(\Delta_b^{(A)}, \Theta_b^{(A)}, \delta T_b^{[\tau]}, \delta x_e^{[\tau]}, \Phi_{\rm GI}, \Psi_{\rm GI}, \ldots\big),$$

and the anomaly output the vector

$$\mathcal{Y}_A = \big(\delta\rho_A, \delta p_A, q_A, \pi_A, \delta Q_A, F_A, \ldots\big).$$

A scalar $B_A(k,a)$ is licensed only when non-density columns are below the declared frozen residual threshold. If geometry feedback is present, the Boltzmann handoff normally receives the bare partial derivatives (E_b, E_g), while the dressed static gain is

$$B_{\rm dressed} = (I - E_g G_A)^{-1}(E_b + E_g G_b).$$

Passing the dressed gain into a solver that separately evolves metric feedback would count gravity twice.

The physical repair rate has the same source discipline. An equilibrium sampler kernel does not determine physical time. The finite parent must provide native rates $q_u(x,y)$ and a physical clock,

$$(L_u f)(x) = \sum_{y\neq x} q_u(x,y)[f(y) - f(x)].$$

The generator descends to the quotient iff, for any two representatives $\pi(x) = \pi(x')$,

$$\sum_{y:\pi(y)=q'} q_u(x,y) = \sum_{y:\pi(y)=q'} q_u(x',y) \quad \text{for every quotient state } q'.$$

Only this native generator, together with quotient lumpability and physical rate units, can promote $\Gamma_{\rm rec}$. Static settled-state response, step response, and the integrated retarded kernel must agree in the common parent model before a one-pole reduction is emitted.

The physical kernel receipts are therefore

```
COMMON_SOURCE_FUNCTIONAL_RECEIPT
ADMISSIBLE_SOURCE_TANGENT_RECEIPT
CONSTRAINT_PRESERVING_RETRACTION_RECEIPT
SOURCE_VECTOR_SUFFICIENCY_RECEIPT
B_A_SOURCE_LIFT_INDEPENDENCE_RECEIPT
SOURCE_DESIGN_IDENTIFIABILITY_RECEIPT
FINITE_DIFFERENCE_ORDER_RECEIPT
C1_REFINEMENT_RECEIPT
ORDER_OF_LIMITS_RECEIPT
NATIVE_REPAIR_GENERATOR_RECEIPT
PHYSICAL_RATE_UNIT_RECEIPT
QUOTIENT_LUMPABILITY_RECEIPT
STATIC_DYNAMIC_RESPONSE_CONSISTENCY_RECEIPT
```

The simulator may emit diagnostic candidates before these receipts pass, but those candidates are not physical B_A, $\rho_{A,\rm eq}$, or $\Gamma_{\rm rec}$ kernels.

13.5 Compressed CMB, BAO, Growth, And S_8 Diagnostic

The checked-in legacy Boltzmann plumbing test passes the anomaly as a cold pressureless component with the flat-benchmark diagnostic parent-grid ratio $\rho_A/\rho_b = 5.363470441$. It uses CAMB 1.6.6, the rejected weighted-cycle comparison value $\sum m_\nu = 0.0900119296\,\text{eV}$, and diagonal Gaussian compressed rows. The neutrino input is a compare-only historical coordinate, not an OPH prediction:

Quantity	Legacy diagnostic	Target	Pull
Planck Ω_m	0.315905207	0.315 ± 0.007	$+0.129$
Planck σ_8	0.807787208	0.811 ± 0.006	-0.535
Planck S_8	0.828924043	0.831027707 ± 0.0111	-0.190
DESI DR1 BAO Ω_m	0.315905207	0.295 ± 0.015	$+1.394$
DESI DR1 BAO/BBN/θ_* H_0	67.4	68.52 ± 0.62	-1.806
Weak-lensing S_8	0.828924043	0.790 ± 0.016	$+2.433$

The Planck rows are in-loop consistency rows, not predictions: the flat-benchmark Ω_A is a Friedmann residual built from measured inputs on a Planck-like budget, fed to CAMB, and scored against Planck. The weak-lensing S_8 row is the one discriminating row in this table; at $+2.433\sigma$ it currently disfavors the benchmark. The historical diagonal compressed value is $\chi^2 = 11.463$ for six rows. Because the run inherited the rejected neutrino input, it is invalidated as a current OPH score and is retained only as a reproducibility record. Even after regeneration with an explicitly declared external neutrino scenario, this is a plumbing check and does not replace Planck, DESI, or weak-lensing likelihoods. A publication-grade calculation needs the OPH $B_A(k,a)$ kernel in a custom Boltzmann module and the full experimental covariances.

14 Likelihood Contracts

14.1 SPARC Disk-Potential Contract

The publication-grade SPARC test must solve the conservative disk equation

$$\nabla^2\Phi = \nabla \cdot \left[\nu_\lambda\left(\frac{|\nabla\Phi_b|}{a_{0,\mathrm{OPH}}}\right)\nabla\Phi_b\right] \tag{96}$$

on an axisymmetric baryonic disk model. The hierarchy must marginalize or profile distance, inclination, disk and bulge mass-to-light ratios, gas scale, intrinsic scatter, velocity covariance, quality cuts, and grid convergence. The unit branch and the $\mathbb{Z}_6$/Poisson branch are fixed before this test; SPARC is used as a comparison, not as the selector for $\lambda_{\mathrm{collar}}$. The equation is scoped to old settled galaxies: its universal extension is excluded by Cassini at 19.22σ, and the contract inherits that scope restriction until a source-derived applicability theorem exists.

14.2 Cluster Map Contract

The cluster forward model uses gas hydrodynamics, stellar and galaxy tracers, transported defect stress, the repair source $\rho_{A,\mathrm{eq}}$, lensing projection, X-ray emissivity, SZ pressure, and covariance-aware map likelihoods. Merging clusters test whether the transported stress lags collisional gas. Relaxed clusters test whether the same stress law returns to the static equilibrium source. The static-only failure mode is part of the likelihood contract.

14.3 Boltzmann Contract

The Boltzmann module exposes

$$\bar{\rho}_A(a), \quad \bar{\rho}_{A,\mathrm{eq}}(a), \quad w_A(a), \quad c^2_{s,A}(k,a), \quad \sigma_A(k,a), \quad Q^\mu_A, \quad B_A(k,a), \quad \Gamma_{\mathrm{rec}}(k,a),$$

with an explicitly declared external neutrino-mass scenario in the same run. The rejected weighted-cycle tuple may appear only as a labeled compare-only scenario; it may not be a default, a source input, or a promotion path. Every dark-sector entry must be derived from a frozen finite covariant parent artifact with no CMB, BAO, lensing, SPARC, or cluster fit in that source step. It must recover the ΛCDM cold-component limit when the exchange and stress corrections are turned off, reproduce the compressed diagnostic rows only as a plumbing check, and then run the full CMB TT/TE/EE, CMB lensing, BAO, supernova, weak-lensing, RSD, and S_8 likelihoods under the same nuisance model used for ΛCDM and $w_0 w_a$. The physical likelihood run is frozen by immutable hashes for the source artifact, Boltzmann solver, and likelihood configuration before the data comparison. The same run must expose official likelihood and CDM-limit reductions at global scope. Shard-local `any()` rollups do not promote the source artifact.

15 Numerical Galaxy Comparison

The OPH capacity benchmark used here is

$$a_{0,\mathrm{OPH}} = 1.029186271 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{97}$$

The value of $a_{0,\mathrm{OPH}}$ is built from the measured H_0 and Λ through the capacity de Sitter branch and the SI clock bridge. Under the strange-loop principle set these are SL-4/SL-5 inputs (`../STRANGE_LOOP_PRINCIPLES` so $a_{0,\mathrm{OPH}}$ inherits two measured cosmological quantities. Every comparison of $a_{0,\mathrm{OPH}}$ or $a_{0,\mathrm{eff}}$ against the empirical acceleration scale in this section is therefore a consistency display between measured inputs, not an OPH prediction. The common empirical reference is

$$a_0^{\mathrm{emp}} = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{98}$$

The $\mathbb{Z}_6$/Poisson branch gives

$$\lambda_{\mathrm{collar}} = 0.934300639, \qquad a_{0,\mathrm{eff}} = 1.179018696 \times 10^{-10}\,\mathrm{m\,s^{-2}}. \tag{99}$$

Quantity	Value	Comparison
$a_{0,\mathrm{OPH}}$	$1.029186271 \times 10^{-10}$	-8.070% versus best same-function RAR
$a_{0,\mathrm{OPH}}$	$1.029186271 \times 10^{-10}$	-14.234% versus empirical reference
$a_{0,\mathrm{eff}}$, $\mathbb{Z}_6$/Poisson	$1.179018696 \times 10^{-10}$	$+5.314\%$ versus best same-function RAR
$a_{0,\mathrm{eff}}$, $\mathbb{Z}_6$/Poisson	$1.179018696 \times 10^{-10}$	-1.748% versus empirical reference

These rows are consistency displays: $a_{0,\mathrm{OPH}}$ carries measured H_0 and Λ, and an exact hit on the common empirical a_0 would require about 13% more protected inactive reserve than the $P/24$ behind $\lambda_{\mathrm{collar}} = e^{-P/24}$ (see the Correction Audit).

15.1 SPARC RAR

At $\lambda_{\mathrm{collar}} = 1$ the response function $\nu_{\mathrm{OPH}}(x)$ of (17) coincides with the McGaugh–Lelli–Schombert empirical interpolating function, which was fitted to the SPARC radial acceleration relation [8]. The SPARC comparisons below therefore score a SPARC-fitted functional form against SPARC itself: they are not held-out tests, and per the sensitivity audit they mostly test normalization rather than shape. Using the public SPARC RAR data [7, 8], the local diagnostic gives:

Scenario	a_0 in $\mathrm{m\,s^{-2}}$	RMS dex	Binned RMS dex
OPH unit	$1.029186271 \times 10^{-10}$	0.134208	0.026169
$\mathbb{Z}_6$/Poisson	$1.179018696 \times 10^{-10}$	0.132834	0.015760
Empirical reference	$1.200000000 \times 10^{-10}$	0.132913	0.016164
Best same-function	$1.119530397 \times 10^{-10}$	0.132947	0.017428

All rows share the SPARC-fitted functional form and are non-held-out; the comparison mostly tests normalization. An exact hit on the common empirical a_0 would require about 13% more protected reserve than $P/24$.

15.2 Rotation Diagnostics

The fixed stellar mass-to-light diagnostic and the first nuisance-profiled diagnostic give:

Scenario	Fixed-M/L RMS	Profiled-M/L χ^2/pt	Profiled RMS
OPH unit	23.366208	1.341948	12.420181
$\mathbb{Z}_6$/Poisson	22.868866	1.390941	12.590203
Empirical reference	22.842145	1.400930	12.624079

Units for the RMS columns are $\mathrm{km\,s^{-1}}$. These rows inherit the SPARC-fitted functional form and the measured-input $a_{0,\mathrm{OPH}}$; they are non-held-out consistency displays that mostly test normalization.

15.3 Systematic Likelihood Scaffold

A stronger local scaffold profiles stellar M/L, distance, inclination, and gas scale per galaxy using the SPARC sample table [6, 7]. It uses $Q \leq 2$, inclination at least 30°, and 3168 rotation-curve rows.

Scenario	χ^2/pt	χ^2/dof	RMS	Median disk M/L
OPH unit	1.016589	1.275672	11.249541	0.519102
$\mathbb{Z}_6$/Poisson	1.021093	1.281324	11.276832	0.497435
Empirical reference	1.023261	1.284045	11.285518	0.493865

The systematic scaffold mildly favors the unit branch. The $\mathbb{Z}_6$/Poisson branch remains the better acceleration-scale and fixed-M/L RAR match. The fixed-M/L improvement alone is weak evidence for the $\mathbb{Z}_6$ bridge theorem. This table shares the non-held-out SPARC provenance and the mostly-normalization sensitivity caveat of the RAR rows above, and the 13% reserve mismatch against the common empirical a_0 applies to its acceleration scales unchanged.

15.4 Audited Fixed-Branch RAR And BTFR Results

The aggregate RAR calculation above is now also evaluated explicitly as a fixed-branch test. Without refitting either OPH branch to the 2,693 public RAR rows from 153 galaxies, the conditional $\mathbb{Z}_6$ scale gives 0.1328335 dex RMS, while the unit endpoint gives 0.1342082 dex. The effective scale that minimizes the same table's diagonal chi-square proxy has RMS 0.1329467 dex. The near equality is an encouraging retrospective check of the fixed formula and scale. It is not a blind prediction: $a_{0,\mathrm{OPH}}$ is a capacity/cosmology benchmark, exact $\lambda_{\mathrm{collar}} = e^{-P/24}$ is conditional, and the aggregate rows share distance, inclination, and stellar-population systematics. A full nuisance-aware SPARC analysis independently reports

$$a_0 = (1.19 \pm 0.04_{\mathrm{stat}} \pm 0.09_{\mathrm{sys}}) \times 10^{-10}\,\mathrm{m\,s^{-2}},$$

placing the conditional $\mathbb{Z}_6$ value near its central result [10].

The BTFR audit corrects an earlier statistics error. Unweighted ordinary least squares discarded the published errors in both M_b and V_f, and the earlier 0.9469-dex headline compared unequal-slope intercepts at $V_f = 1\,\mathrm{km\,s^{-1}}$, far outside the sample. Orthogonal maximum likelihood with both-axis errors and intrinsic perpendicular scatter gives

$$s_{\mathrm{BTFR}} = 3.845654 \pm 0.085817,$$

reproducing the published 3.85 ± 0.09 result [9]. The OPH deep-limit slope 4 is 1.80σ high and lies within the published systematic range 3.5–4.0. At the declared $100\,\mathrm{km\,s^{-1}}$ pivot, however, the fixed-slope fit gives $\log_{10}(M_b/M_\odot) = 9.670792 \pm 0.020827$, whereas the conditional $\mathbb{Z}_6$ normalization predicts 9.805555. The data-minus-model offset is therefore -0.134763 dex, or -6.47σ using only the fit's statistical normalization error. That pull does not yet marginalize a global stellar mass-to-light shift, distances, inclinations, sample selection, or whether every reported V_f has reached the asymptotic regime. The 123-galaxy BTFR table is drawn from SPARC, so it is a separate observable on an overlapping sample rather than an independent galaxy population.

These aggregate checks evaluate the algebraic RAR proxy. They do not replace the conservative disk solve of the likelihood contract. In non-spherical geometry $\nabla\nu \times \mathbf{g}_b$ need not vanish, so a publication-grade OPH test must expose the curl-field residual pattern rather than assume the pointwise algebraic relation.

15.5 Cassini Applicability Stress

If the conservative settled-galaxy equation is extrapolated universally to a Solar source embedded in the measured Galactic external field, it predicts an external-field quadrupole. The audited angular integral reproduces the published spherical benchmark and gives

$$Q_2^{\mathbb{Z}_6} = 3.62018 \times 10^{-26}\,\mathrm{s^{-2}}, \qquad Q_2^{\lambda=1} = 3.40219 \times 10^{-26}\,\mathrm{s^{-2}}.$$

Against the Cassini summary $Q_2 = (1.6 \pm 1.8) \times 10^{-27}\,\mathrm{s^{-2}}$, these are fixed-input residuals of 19.22σ and 18.01σ, respectively [11]. Adding only the linearized Gaia uncertainty on the Galactic acceleration reduces the $\mathbb{Z}_6$ pull to about 10.71σ, still strongly negative [12]. No sign, unit, angular-integration, or branch-mapping error has been found.

This falsifies the natural universal/full-source extension of the displayed static equation. It does not, by itself, falsify the recovered OPH core or the equation's declared old-settled-galaxy scope. The branch now owes a quantitative source-derived applicability, coarse-graining, screening, or transport theorem that preserves the galaxy result while suppressing the Solar-System quadrupole. A free switch introduced after reading Cassini would be a fitted continuation, not that theorem.

15.6 Empirical Implementation Scorecard

The executable empirical pass combines four diagnostic layers: public SPARC tables, the flat-benchmark capacity-saturated homogeneous anomaly state, the finite repair matrix with a cluster timing gate, and the compressed CAMB rows. The checked-in single-pass output below predates the neutrino audit and used the rejected weighted-cycle mass sum. It is an invalidated legacy diagnostic, not a current OPH number surface:

Quantity	Stored legacy diagnostic	Measurement comparison
Ω_A	0.264114401	Planck-like matter budget under $\Omega_K = 0$ benchmark
$\Omega_{\Lambda,\mathrm{OPH}}$	0.684423332	capacity de Sitter branch
ρ_A/ρ_b	5.363470441	flat-benchmark residual ratio
$\gamma_{\text{repair step}}$	0.050717471	finite repair matrix step-gap diagnostic
$a_{0,\mathrm{OPH}}$	$1.029186271 \times 10^{-10}$	8.070% low versus best RAR
$a_{0,\mathrm{eff}}$, $\mathbb{Z}_6$/Poisson	$1.179018696 \times 10^{-10}$	1.748% low versus common empirical a_0
SPARC systematic χ^2/pt, unit	1.016589	best scaffold row
SPARC systematic χ^2/pt, $\mathbb{Z}_6$	1.021093	close scaffold row
CAMB Ω_m	0.315905206	$+0.129\sigma$ versus Planck
CAMB σ_8	0.807787204	-0.535σ versus Planck
CAMB S_8	0.828924037	$+2.433\sigma$ versus weak lensing

Row labels: the CAMB Ω_m and σ_8 rows scored against Planck are in-loop consistency rows, since the flat-benchmark Ω_A residual consumed Planck-like measured inputs before CAMB ran. The weak-lensing S_8 row is the discriminating out-of-loop row and currently disfavors the benchmark at $+2.433\sigma$. The $a_{0,\mathrm{OPH}}$ and $a_{0,\mathrm{eff}}$ rows carry measured H_0 and Λ inside $a_{0,\mathrm{OPH}}$, the non-held-out SPARC provenance of ν_{OPH}, and the 13% reserve mismatch against the common empirical a_0; the SPARC χ^2 rows mostly test normalization.

The stored legacy CMB/BAO/growth score is

$$\chi^2_{\mathrm{diag}} = 11.463 \qquad \text{for six compressed rows.}$$

Within that invalidated run, the pressure point was weak-lensing S_8. The stored value is 0.828924037. The DES/KiDS compressed row is 0.790 ± 0.016.

For the cluster timing gate, take a 200 kpc initial anomaly separation, a 0.2 Gyr time after passage, and an example measured offset 150 ± 50 kpc. The repair matrix gives:

τ_{rec} in Gyr	t_{commit} in Myr	Retained fraction	Offset in kpc	χ^2
0.3	15.2152	0.513417	102.683	0.896
0.5	25.3587	0.670320	134.064	0.102
1.0	50.7175	0.818731	163.746	0.076

This row is a timing gate. Map likelihoods need observed lensing maps, gas maps, stellar maps, merger-age priors, and covariance.

15.7 Implementation Work Packages

The code required for paper-grade empirical tests consists of the following modules:

1. An axisymmetric conservative disk-potential solver for

$$\nabla^2\Phi = \nabla \cdot [\nu(|\nabla\Phi_b|/a_{0,\mathrm{OPH}})\nabla\Phi_b],$$

   with grid-convergence tests and lensing-potential output.

2. A hierarchical SPARC likelihood that samples or profiles distance, inclination, disk M/L, bulge M/L, gas scale, intrinsic scatter, velocity covariance, and selection cuts under one nuisance model.

3. A finite-collar parent evaluator for $\rho_A(a)$, $K_A^{(\rho)}(k,a)$, $B_A(k,a)$, and $\rho_{A,\mathrm{eq}}[X]$, using OPH collar samples as input. The scalar-load diagnostic grid is plumbing only.

4. A CAMB or CLASS anomaly module exposing the background density, stress variables, exchange current, relaxation kernel, environmental response kernel, and an explicitly declared external neutrino-mass scenario in one run. The rejected weighted-cycle tuple may be retained only as a named compare-only scenario and must fail every promotion gate.

5. A cluster forward model with gas hydrodynamics, galaxy and stellar tracers, transported anomaly stress, lensing projection, X-ray and SZ observables, merger-age priors, and map covariance.

6. A likelihood runner for Planck or ACT spectra, CMB lensing, BAO, supernovae, weak lensing, RSD, SPARC, and cluster maps, with model comparison against ΛCDM, empirical MOND functions, halo models, and w_0w_a extensions.

7. A reproducibility harness with fixed data hashes, fixed random seeds, unit checks, and independent numerical replication of the scorecard rows.

15.8 Executable Dark-Sector Simulation Ladder

The simulator counterpart is organized as a promotion ladder, not as a single dark-matter model run. The integration command is

```
python3 -m oph_fpe.cli dark-sector-simulation-plan --run-dir runs/<run_id> \
  --out runs/<run_id>/dark_sector_simulation_plan
```

and emits `DARK_SECTOR_SIMULATION_PLAN_RECEIPT`. The report reads the static galaxy, finite covariant parent, finite-collar Boltzmann bundle, Boltzmann-input, CMB-anomaly, and frozen-likelihood reports, records the first blocked gate, and names the next simulator command. It is a receipt-audit and work-queue object. It is not a physical dark-sector prediction, not a fitted likelihood, and not a substitute for source-generation receipts.

The gate order is:

1. Static galaxy diagnostic: `run-galaxy-static` must emit `STATIC_GALAXY_RAR_BTFR_RECEIPT`. This checks only the settled RAR/BTFR bridge.

2. Finite covariant parent: `finite-covariant-collar-parent` must emit `FINITE_COVARIANT_COLLAR_PACKET_` This is the covariant source artifact required before physical stress and exchange-current claims can be promoted.

3. Finite-collar Boltzmann source bundle: `finite-collar-boltzmann-bundle` must emit the diagnostic source-bundle receipt and, for physical promotion, `PHYSICAL_BOLTZMANN_EXPORT_CERTIFICATE`.

4. Boltzmann input bridge: `oph-boltzmann-inputs` must expose the CDM-limit solver row and keep diagnostic repair-exchange rows separate from physical inputs.

5. Frozen transfer and likelihood closure: `frozen-transfer-likelihood` must certify source freeze, solver pins, CDM-limit regression, Standard-Model-off controls, blinded setup, and official likelihood execution before the dark branch can claim a likelihood-evaluated physical prediction.

The planner also records optional `cmb-anomaly-report` output. That diagnostic can guide anomaly studies. The physical dark-matter promotion gate uses it only with passing finite-parent, Boltzmann, and likelihood receipts.

15.9 Correction Audit

The comparison mostly tests normalization because the OPH static law uses the same acceleration-family shape as the empirical RAR fit. If $a_{0,\mathrm{OPH}}$ is held fixed, different targets imply different $\lambda_{\mathrm{collar}}$ values:

Target	Required $\lambda_{\mathrm{collar}}$	Reserve $-\ln\lambda_{\mathrm{collar}}$	Ratio to $P/24$
Best same-function RAR	0.958802256	0.042070424	0.619074
Binned RAR	0.936737944	0.065351711	0.961663
Common empirical a_0	0.926096769	0.076776547	1.129781
Fixed-M/L rotation	0.926654674	0.076174302	1.120919
OPH $\mathbb{Z}_6$/Poisson	0.934300639	0.067957009	1.000000

This rules out several tempting shortcuts. In the normalization-profiled all-point objective, a generalized activation exponent near 0.503420 appears. With fixed $a_{0,\mathrm{OPH}}$, direct RAR objectives give exponents near 0.516 all-point and 0.519 weighted-bin. These shifts are objective-dependent, and flat deep-IR rotation requires 0.5. Moving the exponent would buy little and would damage the BTFR theorem.

Finite-thickness nonuniformity can raise $\lambda_{\mathrm{collar}}$ above $\exp(-P/24)$, which moves the coefficient toward the binned RAR value. It cannot lower $\lambda_{\mathrm{collar}}$ toward the common empirical or fixed-M/L values while the mean reserve remains $P/24$ and Poisson counting is retained. An exact hit to the common empirical acceleration would require about 13% more protected inactive reserve than $P/24$, or a different microscopic selector. That extra reserve needs an OPH theorem before use.

16 Claim Boundaries

Pressure point	OPH route	Status
Unknown particle	Dark source is a carried modular/collar information-defect stress.	Conditional source target.
Galaxy RAR and BTFR	Collar/cut activation gives the square-root transition and Poisson activation law under independent-increment repair counting; universal extension excluded by Cassini at 19.22σ.	Conditional static theorem, old-settled-galaxy scope only.

Pressure point	OPH route	Status
Acceleration scale	Capacity scale gives $a_{0,\mathrm{OPH}}$ from measured H_0/Λ (SL-4/SL-5 inputs); $\mathbb{Z}_6$/Poisson reserve gives $a_{0,\mathrm{eff}}$ close to the empirical scale, 13% short in reserve of an exact hit.	Conditional coefficient theorem; in-loop consistency display.
Galaxy lensing	No-slip branch gives $\Phi = \Psi$ if the information-defect stress is dust on settled galaxies.	Conditional stress theorem.
Cluster offsets	Transported anomaly can lag collisional gas; the Jacobi clock gives a no-fit scale under the tidal-mismatch premise.	Forward likelihood contract.
CMB and growth	Flat capacity state selection gives a CDM-like homogeneous residual; the closed branch has a conserved load current; environmental kernels describe nonzero-field response.	Amplitude selector and real Boltzmann likelihood are not emitted by this paper.

The supported claim is:

> OPH defines an imperfect-information correction to the recovered Einstein branch. Under explicit bridge premises, the static galaxy equation reproduces the RAR/BTFR functional form from collar/cut repair statistics and yields an acceleration scale close to SPARC as an in-loop consistency display, since $a_{0,\mathrm{OPH}}$ carries measured H_0 and Λ as SL-4/SL-5 inputs. The unit branch is low in normalization. The $\mathbb{Z}_6$/Poisson branch lands within 1.748% of the common empirical acceleration reference. Profiled stellar, distance, inclination, and gas nuisances mildly favor the unit branch. The Jacobi clock gives a no-fit cluster-offset scale under its tidal-mismatch premise. Flat capacity state selection gives a Planck-like homogeneous information-defect residual, while closed-branch transport gives a conserved load current. Cluster and CMB/growth likelihoods require measured cluster covariances, a finite-screen amplitude selector, a finite covariant source parent, recipient stress and exchange-current closure for any nonzero repair exchange, physical-clock receipts for any promoted Γ_{rec}, and a frozen Boltzmann/likelihood implementation.

Claim outside this paper:

> OPH has fully derived dark matter and solved MOND's cluster and CMB problems.

That stronger statement requires SPARC publication likelihoods, cluster likelihoods, a finite covariant parent for the cosmological source functions, and frozen Boltzmann likelihoods.

17 Proof Ledger

Target	OPH closure	Numerical state
Galaxy source	Minimal scalar recoverability defect $R_C = I(A : D\|B)$.	Conditional source target.
Quotient-edge locality	Scalar repair opportunities are edge-center cut-register events on the physical quotient.	Conditional bridge.
Coherent material source lift	A nondegenerate OPH-coherent material source $S^{\mathrm{can}}_{\mathrm{coh}}$ descends to the quotient and is forced into the same edge-center scalar channel by Scalar Edge-Center Exhaustion.	Theorem-grade scalar-channel identity in the screen microphysics and χ_ν papers; finite stress-parent, abundance, and Boltzmann gates are separate obligations.
Finite channel bridge	One finite channel carries both the scalar-slot record panel and the collar-slice reserve panel, so the source generator and collar survival factor are composed on the same indexed family.	Bookkeeping bridge; nature-instantiation and G9 numerical calibration remain open physical obligations.
Finite thickness	Uniform product-thickening plus slice-wise scalar-reserve unbiasedness licenses the exact coefficient; otherwise the profile integral and, when scalar-weighted mean passes, Jensen band apply.	Conditional imported theorem.
No-slip stress	Positive kinetic collar measure; settled monokinetic branch has dust stress and vanishing information-sector anisotropic stress.	Conditional stress theorem.
Repair relaxation	Fixed repair-commit chain gives $\tau_{\mathrm{rec}} = t_{\mathrm{commit}}/\gamma_{\log}$.	Conditional packet theorem and clock candidate.
CMB and growth	Nonzero-environment kernel is explicit; exact FLRW static RAR kernel is blocked by a no-go theorem.	Finite-parent Boltzmann contract.
Abundance history	Flat capacity state selection gives a CDM-like residual; the closed transported branch has a conserved load current and charge.	Amplitude selector not emitted by this paper.
Linear transfer	The perturbation equation has a relaxation transfer solution between CDM-like and tracking limits.	Requires evaluated $B_A(k, a)$, recipient stress and exchange-current closure for nonzero exchange, physical-clock receipts for any promoted Γ_{rec}, and frozen hashes.
Parent functional	The finite collar expectation of the positive repair defect emits $\rho_{A,\mathrm{eq}}$ and B_A if its FLRW limit is regular.	Scalar source target, not full stress closure.
Repair matrix	A reversible Metropolis kernel on packet states gives a concrete K.	Conditional on declared packet schema.

17.1 Positive Galaxy Source

Theorem target 0A. The scalar carried charge is fixed by the exact modular-defect identity, not by a separate uniqueness principle. For a faithful finite collar state ω_{ABD},

$$-\langle K_{ABD} - K_{AB} - K_{BD} + K_B\rangle_\omega = I_\omega(A : D|B).$$

After reference subtraction on an exact Markov collar, this defines the canonical carried modular charge $M_C = I_\omega(A : D|B)$ in nats. A quotient-invariant collar allocation may sum these charges over a proper small-ball cover only when it passes the source-localization saturation receipt connecting the same carried operator to the finite-cutoff Einstein remainder.

Explicit premise. The galaxy-scale finite-stage modular nonadditivity is the carried modular charge appearing in the fixed-cap small-ball remainder, with hidden carrier duplicates, port relabelings, repair schedules, and refinement-equivalent covers quotiented out.

Conditional argument. Conditional mutual information is nonnegative, so the source-localized carried charge is nonnegative. Equation (7) then gives a nonnegative rest-mass density after the source-localization receipt passes. The scalar charge alone does not determine T_A^{ab}; packet factorization, causal directions, allocation, rest frame, and transport law are required to lift it to a stress tensor. □

Proof obligation. Prove the operator-level source-localization relation between the finite-stage small-ball modular remainder and the quotient-invariant collar charge, then prove the positive packet allocation and covariant causal packet lift.

Failure consequence. Without source localization, $I(A : D|B)$ is only a recoverability and collar-diagnostic number. Without the packet lift, it cannot support lensing, CMB, growth, or conservation claims as a physical stress source.

17.2 Quotient-Edge Scalar Opportunity

Imported theorem. The fixed-cutoff uniqueness statement is discharged in the microphysics paper by

$$\mathsf{Scal}_{r,C} = \mathcal{E}_{r,C}$$

on the edge-center scalar-completeness branch, and by the following imports:

$$\Pi^{\text{scal}}_{r,C} \in \mathcal{E}_{r,C}, \qquad Z_{6,r,C} \in \mathcal{E}_{r,C}, \qquad [\Pi^{\text{scal}}_{r,C}, Z_{6,r,C}] = 0.$$

The relevant theorem labels in the microphysics paper are

`thm:scalar-activation-edge-center`, `thm:scalar-z6-same-register-commute`, `thm:scalar-channel-ex`

For coherent material support the imported source is

$$S^{\text{can}}_{\text{coh}} = \mathbf{1}_{\text{self-read}}\, \mathsf{R}_U\, \mathsf{P}_U\, \mathsf{C}_U \in \mathcal{E}_{r,C}$$

on the nondegenerate branch. This import fixes the scalar channel used by the χ_ν coefficient and does not supply a finite covariant dark-stress parent.

Branch boundary. The import is conditional on the microphysics assumptions labeled `ass:edge-center-scala` and `ass:scalar-activation-channel-completeness`. If either fails, then scalar activation may require an additional quotient-visible scalar carrier, and the $\mathbb{Z}_6$ reserve cannot determine λ_{collar} without a new co-registration theorem.

17.3 Uniform Finite-Thickness Reserve

Correct exact-value premise. The exact finite-thickness coefficient requires the uniform product-thickening branch of the microphysics paper:

$$\mathcal{A}^{\text{thick}}_{r,C} \cong \mathcal{E}_{r,C} \otimes \mathcal{T}_r(Y),$$

$$\tau^{\text{thick}}_{q,r,C} = \tau^{\mathcal{E}}_{q,r,C} \otimes \tau^{Y}_{r},$$

$$Z^{\text{thick}}_{6,r,C} = Z_{6,r,C} \otimes 1_Y,$$

and, essentially,

$$\frac{\tau^{\mathcal{E}}_{q,r,C}\left(\Pi^{\text{scal}}_{r,C}(y) Z_{6,r,C}\right)}{\tau^{\mathcal{E}}_{q,r,C}\left(\Pi^{\text{scal}}_{r,C}(y)\right)} = \frac{P}{24}$$

for every scalar-active transverse slice y. Product trace alone is not enough. The scalar-active sampler must be reserve-unbiased.

Imported theorem. With local Poisson reserve survival and the uniform product-thickening branch, the microphysics theorem labeled `thm:exact-uniform-product-thickening-coefficient` gives

$$\lambda_{\text{collar}} = e^{-P/24} = 0.9343006394893864\ldots.$$

Without the uniform branch, the theorem-grade coefficient is

$$\lambda_{\text{collar}} = \int \mathrm{d}y\, w(y) e^{-\epsilon_{\mathbb{Z}_6}(y)}.$$

If the scalar-weighted protected reserve mean is $P/24$, Jensen gives only

$$e^{-P/24} \leq \lambda_{\text{collar}} \leq 1.$$

17.4 No-Slip Settled Stress

Theorem target 0D. Let the information-defect stress be the second moment of a positive finite-collar kinetic measure $\mathrm{d}\mathcal{E}_x(v)$ on the future unit timelike shell:

$$T^{ab}_{\mathcal{I}} = \int v^a v^b \, \mathrm{d}\mathcal{E}_x(v). \tag{100}$$

If the settled galaxy branch is monokinetic in its rest frame, then

$$T^{ab}_{\mathcal{I}} = \rho_{\mathcal{I}} u^a_{\mathcal{I}} u^b_{\mathcal{I}}, \qquad P_{\mathcal{I}} = 0, \qquad q^a_{\mathcal{I}} = 0, \qquad \pi^{ab}_{\mathcal{I}} = 0. \tag{101}$$

For negligible or separately modeled anisotropic stress in all other sectors, $\Phi = \Psi$.

Explicit premise. The finite collar packet state emits the positive kinetic measure, and the settled galaxy branch is monokinetic at the coarse scale used for lensing.

Conditional argument. A positive measure on future timelike velocities has a symmetric future-positive second moment. In the monokinetic branch the measure is concentrated at one four-velocity, so the second moment is $\rho_{\mathcal{I}} u^a_{\mathcal{I}} u^b_{\mathcal{I}}$. Its pressure, heat flux, and anisotropic stress vanish in that rest frame. The off-diagonal spatial Einstein equation then gives $\Phi = \Psi$ when the other sectors carry no uncompensated anisotropic stress. □

Proof obligation. Derive $\mathrm{d}\mathcal{E}_x(v)$ from the finite OPH collar packet state or from an equivalent covariant parent action.

Failure consequence. If pressure or anisotropic stress survives coarse graining, the lensing slip equation must include those terms and no-slip is regime-dependent.

17.5 Repair Relaxation Scale

Theorem 0E. Let K be the fixed-cutoff repair-commit transition matrix for one local coarse cell at fixed baryonic source. If K is finite, irreducible, and aperiodic, with largest nonstationary eigenvalue $r_\star$ in modulus, then

$$\tau_{\mathrm{rec}} = \frac{t_{\mathrm{commit}}}{-\log|r_\star|}. \tag{102}$$

If every allowed repair move has probability at least $q_{\min} > 0$ and the comparison chain has conductance Φ_K, then

$$1 - |r_\star| \geq \frac{1}{2} q_{\min} \Phi_K^2. \tag{103}$$

Proof. A finite irreducible aperiodic chain has a unique stationary state. Every nonstationary mode decays as $|r_j|^n$ in repair-step time. One physical commit step takes t_{commit}, so physical relaxation has rate $-\log|r_\star|/t_{\mathrm{commit}}$. The conductance bound follows from the standard Cheeger comparison for the reversible comparison chain and the lower move probability $q_{\min}$. □

Theorem 0E'. Equilibrium data do not select physical repair kinetics. Let P be an irreducible finite transition matrix with stationary law $\pi P = \pi$. For any $0 < \alpha \leq 1$, define

$$P_\alpha = (1-\alpha)I + \alpha P.$$

Then $\pi P_\alpha = \pi$. If r_j is an eigenvalue of P, the corresponding eigenvalue of P_α is

$$r_j(\alpha) = 1 - \alpha + \alpha r_j,$$

and the physical decay rate at commit time $\Delta t_{\mathrm{physical}}$ is

$$\Gamma_j(\alpha) = -\frac{\log|1-\alpha+\alpha r_j|}{\Delta t_{\mathrm{physical}}}.$$

Thus the same equilibrium law, $\rho_{A,\mathrm{eq}}$, B_A, and detailed balance can support different kinetics. A sampler built from an equilibrium functional is a valid equilibrium evaluator only after its proposal law, attempt frequency, causal neighborhood, active fiber, and physical clock are declared as part of the OPH repair generator. □

17.6 Environmental Kernel Around A Nonzero Field

Conditional theorem 0F. Let the settled anomaly field be

$$\mathbf{g}_A = (\nu_{\rm OPH} - 1)\mathbf{g}_b. \tag{104}$$

Linearize around a nonzero background field $\bar{\mathbf{g}}_b$. With

$$x = |\bar{\mathbf{g}}_b|/a_{0,\rm OPH}, \qquad n_i = \bar{g}_{b,i}/|\bar{\mathbf{g}}_b|,$$

the Fourier density response to a baryonic density mode is

$$K_A^{(\rho)}(x,\mu) = \nu_{\rm OPH}(x) - 1 + x\nu'_{\rm OPH}(x)\mu^2, \tag{105}$$

where μ is the cosine between the mode direction and $\mathbf{n}$. The contrast kernel is

$$B_A(k,a) = \frac{\bar{\rho}_b(a)}{\bar{\rho}_{A,\rm eq}(a)} K_A^{(\rho)}(k,a). \tag{106}$$

Proof. The first variation is

$$\delta g_{A,i} = \big([\nu_{\rm OPH}(x) - 1]\delta_{ij} + x\nu'_{\rm OPH}(x)n_i n_j\big)\,\delta g_{b,j}. \tag{107}$$

Taking the divergence and using the Fourier-space Poisson relation for $\delta\mathbf{g}_b$ gives the displayed scalar multiplier. The derivative of

$$\nu_{\rm OPH}(x) = [1 - \exp(-\lambda_{\rm collar}\sqrt{x})]^{-1}$$

is

$$\nu'_{\rm OPH}(x) = -\frac{\lambda_{\rm collar}\exp[-\lambda_{\rm collar}\sqrt{x}]}{2\sqrt{x}[1 - \exp(-\lambda_{\rm collar}\sqrt{x})]^2}.$$

□

17.7 Background Abundance Green Function

Theorem 0G. Let the background anomaly density obey

$$\rho'_A + 3\mathcal{H}\rho_A = -a\Gamma_{\rm rec}(\rho_A - \rho_{A,\rm eq}). \tag{108}$$

Define the comoving densities

$$R_A = a^3\rho_A, \qquad R_{A,\rm eq} = a^3\rho_{A,\rm eq}. \tag{109}$$

Then

$$R_A(\eta) = W(\eta,\eta_i)R_A(\eta_i) + \int_{\eta_i}^{\eta} {\rm d}\eta'\, W(\eta,\eta')a(\eta')\Gamma_{\rm rec}(\eta')R_{A,\rm eq}(\eta'), \tag{110}$$

where

$$W(\eta,\eta') = \exp\left[-\int_{\eta'}^{\eta} {\rm d}\tilde{\eta}\, a(\tilde{\eta})\Gamma_{\rm rec}(\tilde{\eta})\right]. \tag{111}$$

Proof. Multiplying the background equation by a^3 gives

$$R'_A = -a\Gamma_{\rm rec}(R_A - R_{A,\rm eq}). \tag{112}$$

This is a first-order linear equation. Multiplication by the integrating factor

$$\exp\left[\int_{\eta_i}^{\eta} \mathrm{d}\eta'\, a(\eta')\Gamma_{\rm rec}(\eta')\right] \tag{113}$$

and integration gives the stated expression. □

The theorem shows exactly where abundance information enters: the initial comoving anomaly load, the equilibrium source, and the repair rate. These are OPH microphysical outputs, not SPARC parameters.

17.8 Linear Relaxation Transfer

Theorem 0H. For fixed k, assume the no-slip anomaly perturbation equation

$$\delta'_A = -\theta_A + 3\Phi' - a\Gamma_{\rm rec}q_A(\delta_A - B_A\delta_b). \tag{114}$$

Define

$$\Xi_A = a\Gamma_{\rm rec}q_A. \tag{115}$$

Then

$$\begin{aligned}\delta_A(\eta,k) = {} & W_A(\eta,\eta_i)\delta_A(\eta_i,k) \\ & + \int_{\eta_i}^{\eta} \mathrm{d}\eta'\, W_A(\eta,\eta')\left[-\theta_A + 3\Phi' + \Xi_A B_A\delta_b\right]_{\eta',k},\end{aligned} \tag{116}$$

with

$$W_A(\eta,\eta') = \exp\left[-\int_{\eta'}^{\eta} \mathrm{d}\tilde\eta\, \Xi_A(\tilde\eta,k)\right]. \tag{117}$$

Proof. Move the damping term to the left:

$$\delta'_A + \Xi_A\delta_A = -\theta_A + 3\Phi' + \Xi_A B_A\delta_b. \tag{118}$$

The integrating-factor solution is the displayed expression. □

This formula makes the two useful limits explicit. If $\Xi_A/\mathcal{H} \ll 1$, the anomaly evolves like pressureless free-falling matter. If $\Xi_A/\mathcal{H} \gg 1$, it tracks the OPH equilibrium source up to derivative corrections. Cluster offsets and settled galaxies require the same relaxation operator to sit between those limits.

17.9 Static-To-Linear Separation

Theorem 0I. The nonlinear static RAR law

$$g_{\rm obs} = \frac{g_b}{1 - \exp[-\lambda_{\rm collar}\sqrt{g_b/a_{0,\rm OPH}}]} \tag{119}$$

does not define the FLRW anomaly abundance or the linear CMB kernel by itself.

Proof. The RAR law is written for a static weak-field acceleration vector sourced by a resolved baryonic inhomogeneity. An exactly homogeneous FLRW background has no preferred spatial acceleration vector. Its scalar data are background densities and perturbation kernels, with no radial acceleration profile. In addition, the RAR map contains $\sqrt{g_b}$, so its first derivative is singular at $g_b = 0$ if treated as a homogeneous-background Taylor series. Therefore the static law cannot be linearized into CMB equations without an OPH parent functional. A scalar parent functional must emit $\rho_A(a)$ and $B_A(k,a)$, which belongs to the source-candidate layer. Physical Boltzmann promotion requires the finite covariant packet parent above, its source-provenance certificate, and pooled global reducers. □

17.10 Parent Collar Functional

Scalar source target 0J. Let $\mathcal{C}_x(a)$ be the finite collar family contributing to a coarse FLRW cell x at scale factor a. Assume this finite-collar state selection has a regular FLRW limit. Let $\omega[\rho_b]$ be the OPH collar state selected by the baryonic source, and let $R_C^{\rm src}$ be the typed source routed by the source-route theorem:

$$R_C^{\rm src}[\omega] = \begin{cases} \delta\langle K_C^{\rm anom}[\sigma]\rangle, & \texttt{FIXED\_REFERENCE\_MODULAR\_ENERGY}, \\ I_\omega(A:D|B), & \texttt{NONLINEAR\_CMI\_STRESS} \text{ with matching receipt.} \end{cases} \tag{120}$$

Define

$$\rho_{A,\rm eq}(x,a)c^2 = \frac{15}{8\pi^2\ell(a)^4}\int_{\mathcal{C}_x(a)} \mathrm{d}\mu_C\, R_C^{\rm src}[\omega[\rho_b]]. \tag{121}$$

Then (121) is a scalar parent collar functional that emits a background equilibrium source and a linear response operator:

$$\bar\rho_{A,\rm eq}(a) = \rho_{A,\rm eq}[\bar\rho_b](a), \tag{122}$$

$$(K_{A,r}^{(\rho)})_{jj'}(a) = \left\langle v_{rj}, \left.\frac{\delta\rho_{A,\rm eq}(x,a)}{\delta\rho_b(x',a)}\right|_{\bar\rho_b} v_{rj'}\right\rangle, \qquad (B_{A,r})_{jj'}(a) = \frac{\rho_b(a)}{\rho_{A,\rm eq}(a)}(K_{A,r}^{(\rho)})_{jj'}(a). \tag{123}$$

Conditional argument. The collar measure, the conditional mutual information, and the selected collar state are quotient-local scalar data. The integral is therefore a scalar density on the coarse FLRW cell. Evaluating the functional on a homogeneous baryonic density gives the background equilibrium source. Taking the first functional derivative around that homogeneous state gives a translation-invariant and rotation-invariant linear response kernel. Its finite mode-basis matrix is $K_{A,r}^{(\rho)}(a)$, and division by the equilibrium background density gives $B_{A,r}(a)$. The scalar $B_A(k,a)$ is the diagonal, mode-decoupled special case after the commutator and off-diagonal leakage receipts pass. □

The scalar reduction is accepted only on the physical projectors supplied by the scale bridge. In projector notation,

$$K_{A,r;IJ}^{(\rho)}(a) = P_{r,I}\mathcal{K}_{A,r}^{(\rho)}(a)P_{r,J}, \qquad B_{A,r;IJ}(a) = \frac{\bar\rho_b(a)}{\bar\rho_{A,\rm eq}(a)}K_{A,r;IJ}^{(\rho)}(a).$$

Reporting a scalar $B_A(k,a)$ requires

$$\epsilon_{\rm off}(a) = \frac{\left(\sum_{I\neq J}|P_{r,I}\mathcal{K}_A P_{r,J}|^2\right)^{1/2}}{|\mathcal{K}_A| + \epsilon_0} \le \bar\epsilon_{\rm off}$$

and, inside every degenerate projector,

$$P_{r,I}\mathcal{K}_A P_{r,I} = b_I(a)P_{r,I} + E_I, \qquad \frac{|E_I|}{|b_I| + \epsilon_0} \le \bar{\epsilon}_{\rm iso}.$$

The evidence bundle records `mode_basis_hash`, `background_hash`, `clock_hash`, and `scale_bridge_hash`. $\rho_A(a)$, $B_A(k,a)$, and $\Gamma_{\rm rec}(k,a)$ must share one frozen geometry and clock generation. A transition matrix eigenvalue per repair cycle is a diagnostic until a physical interval $\Delta\tau_r$, an active response block, and either a spectral-radius formula or a Lyapunov/semigroup bound are certified. A scalar $B_A(k,a)$ fails when off-diagonal mode mixing exceeds its frozen tolerance or when the result depends on unitary rotations inside a degenerate physical projector.

The finite-collar evaluation of (121) is the mathematical target for replacing a fitted dark-matter abundance, but it does not determine the full fluid closure. The scalar parent leaves open w_A, $c^2_{s,A}$, σ_A, Q^μ_A, exchange-current closure, a certified $\Gamma_{\rm rec}$, recipient stress, local-frame/quotient covariance, gauge-invariant perturbations, and causal response. Those data must come from the finite covariant packet parent before the scalar tables can feed a physical Boltzmann module.

Proof obligation. Evaluate the collar measure and selected state from finite OPH screen/collar data in the homogeneous, perturbative, galaxy, and cluster regimes; lift the result to a finite covariant packet parent; prove exact stress-energy closure, gauge independence of $B_{A,r}$, common mode-basis compatibility with the primordial source, physical-time calibration for $\Gamma_{\rm rec}$, recipient stress for nonzero exchange, exchange-current closure, local-frame and carrier-quotient invariance, causal/stable response, refinement convergence, and CDM-limit recovery.

17.11 Finite Repair Matrix

Theorem 0K. Let S be the finite packet-state space declared by one fixed-cutoff overlap interface. Let $\pi_{\rm eq}(s|\rho_b)$ be the equilibrium packet distribution emitted by the parent collar functional. Let $q(s,s')$ be an inverse-closed local proposal rule generated by the declared repair menu, with $q(s,s') > 0$ iff $q(s',s) > 0$. Define, for $s \neq s'$,

$$K(s,s') = q(s,s') \min\left(1, \frac{\pi_{\rm eq}(s'|\rho_b)q(s',s)}{\pi_{\rm eq}(s|\rho_b)q(s,s')}\right), \tag{124}$$

and

$$K(s,s) = 1 - \sum_{s' \neq s} K(s,s'). \tag{125}$$

Then K is a finite stochastic repair transition matrix with stationary distribution $\pi_{\rm eq}$. If the proposal graph is connected and some rejection or hold probability is nonzero, K is irreducible and aperiodic.

Proof. Each row sums to one by construction, so K is stochastic. For $s \neq s'$, inverse-closed proposals make both directions available. Hence

$$\pi_{\rm eq}(s)K(s,s') = \min\bigl(\pi_{\rm eq}(s)q(s,s'), \pi_{\rm eq}(s')q(s',s)\bigr) = \pi_{\rm eq}(s')K(s',s). \tag{126}$$

Detailed balance proves stationarity. Connected proposal support gives irreducibility, and a nonzero hold probability gives aperiodicity. □

This construction declares K from OPH data: packet states, local proposals, and the parent equilibrium distribution.

Scalar-load diagnostic. The executable packet diagnostic takes

$$\mathsf{S} = \{0, 1, \ldots, 40\}, \qquad \pi_{\mathrm{eq}}(n) = \mathrm{Pois}(n; 5.363470441)$$

with truncation at 40, nearest-neighbor inverse-closed proposals, and hold probability 0.25. The resulting matrix has row-sum error 1.110×10^{-16}, detailed-balance error 6.939×10^{-18}, and

$$\gamma_{\text{repair step}} = 0.050717471.$$

This diagnostic is a finite packet schema. The OPH task is to derive that schema, or its replacement, from the finite collar states and then attach the stress, recipient, response, convergence, and frozen-likelihood receipts needed for physical transfer.

17.12 Finite Repair-Graph Gap

Theorem 0L. For a declared fixed-cutoff repair graph with transition matrix K, the relaxation time used by clusters and perturbations is computable from the second eigenvalue. If K is reversible with stationary distribution π_{eq}, then

$$r_\star = \max_{\lambda_j \neq 1} |\lambda_j(K)|, \qquad g_{\mathrm{abs}} = 1 - r_\star \tag{127}$$

in discrete repair-step units. If the commit interval is $\Delta t_{\mathrm{commit}}$, the exact physical e-folding rate is

$$\Gamma_{\mathrm{phys}} = -\frac{\log r_\star}{\Delta t_{\mathrm{commit}}}, \qquad \tau_{\mathrm{phys}} = \frac{\Delta t_{\mathrm{commit}}}{-\log r_\star}. \tag{128}$$

The expression $\Delta t_{\mathrm{commit}}/(1 - r_\star)$ is only the small-gap approximation. For continuous-time generator L, the rate is the smallest positive eigenvalue of $-L$ on the mean-zero subspace. On a nonequilibrium nonnormal branch, eigenvalues alone are not enough; the parent must instead provide a Lyapunov, singular-value, or logarithmic-norm bound that excludes transient amplification.

Proof. Reversibility makes K self-adjoint in $L^2(\pi_{\mathrm{eq}})$. The stationary vector has eigenvalue one. Every mean-zero mode decomposes into eigenvectors and decays by its eigenvalue per repair step. The slowest decaying nonstationary mode is therefore the largest nontrivial absolute eigenvalue $r_\star$. After $n = t/\Delta t_{\mathrm{commit}}$ commits, its amplitude is $r_\star^n = \exp[-t(-\log r_\star)/\Delta t_{\mathrm{commit}}]$, which gives the stated physical rate. The continuous-time statement follows by exponentiating the generator. □

This is an exact finite-matrix target for the repair/commit microphysics. A cluster timescale is a prediction only after this matrix or its OPH state functional is supplied.

17.13 Jacobi Repair Clock

Theorem target 0M. Assume repair timing is set by the overlap-visible Jacobi mismatch rate of transported information-defect records. Let $E_{ij} = C_{0i0j}$ be the electric Weyl tensor in the local free-fall frame, equivalently the trace-free Newtonian tidal tensor in the weak-field limit. Homogeneous FLRW Ricci focusing is excluded from this local cluster-offset clock. The scalar repair rate is

$$\Gamma_J = \left(\frac{E_{ij} E^{ij}}{6} \right)^{1/4}. \tag{129}$$

For a Newtonian point source this gives

$$\Gamma_J = \sqrt{\frac{GM}{r^3}}, \qquad \tau_J = \sqrt{\frac{r^3}{GM}}. \tag{130}$$

Explicit premise. The repair-cycle clock is controlled by overlap-visible geodesic-deviation mismatch of transported information-defect records.

Conditional argument. The relative acceleration of neighboring transported records is controlled by the Jacobi operator. The local cluster-offset part is the trace-free Weyl electric tensor E_{ij}; the homogeneous FLRW Ricci term changes the background expansion and is excluded from the local offset between collisional gas and transported record stress. A scalar clock made from E_{ij}, with no chosen direction and no extra dimensional constant, must be a power of $E_{ij}E^{ij}$. The fourth root has units of inverse time. For a point Newtonian source the eigenvalues of E_{ij} are $-2GM/r^3, GM/r^3, GM/r^3$ up to sign convention, hence $E_{ij}E^{ij} = 6(GM/r^3)^2$. The normalization in the theorem makes the point-source result exact. □

Proof obligation. Derive from OPH packet microphysics that the commit clock is the overlap-visible Weyl-Jacobi mismatch clock, while pixel, de Sitter, entropy-production, local-acceleration, and Ricci-curvature clocks are excluded. The self-consistent prediction evaluates E_{ij} from the total weak-field potential determined by baryons plus information-defect stress.

Failure consequence. Without that derivation, Γ_J is a no-fit dynamic diagnostic rather than a repair-clock theorem.

Static-clock boundary. The static equilibrium equation contains $\rho_{A,\mathrm{eq}}$ but no relaxation rate. Replacing Γ_{rec} by any positive function leaves the same settled solution. The Jacobi clock is therefore a dynamic OPH premise and has no theorem claim inside the static RAR law.

17.14 Homogeneous Anomaly State Selection

Theorem target 0N. Under capacity-saturated FLRW state selection, the homogeneous closure relation is

$$\Omega_{A0} + \Omega_{K0} = 1 - \Omega_{\Lambda,\mathrm{OPH}} - \Omega_{b0} - \Omega_{\nu 0} - \Omega_{r0}, \qquad \Omega_{\Lambda,\mathrm{OPH}} = \left(\frac{H_{\mathrm{dS}}}{H_0}\right)^2. \tag{131}$$

If $\Omega_{K0} = 0$ is separately selected or explicitly assumed, this reduces to the flat benchmark value for Ω_{A0}. Otherwise it is only a sum rule. If the homogeneous anomaly charge is transported as dust with no homogeneous repair exchange, then

$$\rho_A(a) = \rho_{A0} a^{-3}. \tag{132}$$

Explicit premise. The homogeneous state is capacity-saturated. The flat version additionally assumes or derives $\Omega_{K0} = 0$. A residual Friedmann constraint after H_0, baryons, neutrinos, and radiation are supplied does not by itself select either the curvature sector or the anomaly abundance.

Conditional argument. The flat Friedmann equation at $a = 1$ is

$$1 = \Omega_{\Lambda,\mathrm{OPH}} + \Omega_{b0} + \Omega_{\nu 0} + \Omega_{r0} + \Omega_{A0} + \Omega_{K0}. \tag{133}$$

Solving gives only $\Omega_{A0}+\Omega_{K0}$. Pressureless homogeneous transport with zero homogeneous exchange obeys

$$\rho_A' + 3\mathcal{H}\rho_A = 0,$$

whose solution is $\rho_A(a) = \rho_{A0} a^{-3}$. □

Proof obligation. Derive the conserved homogeneous information-defect charge Q_A by the source-only anomaly abundance selector of Section 17.15 and supply an independent curvature label, or declare the abundance as cosmological state data.

Failure consequence. The CMB dark abundance is conditional when Q_A lacks `RHO_A_SOURCE_RECEIPT`. A nonzero exchange branch is also conditional unless the same finite parent exposes the recipient stress and equal-and-opposite exchange current that balances the exchange.

Capacity boundary. The screen capacity fixes the dimensionless de Sitter capacity relation; the capacity is the SL-4 estimate read from measured Λ (`../STRANGE_LOOP_PRINCIPLES.md`), and the SI de Sitter scale additionally uses the selected OPH scale certificate. After H_0 and that scale display are supplied, it fixes $\Omega_{\Lambda,\mathrm{OPH}}$; Λ itself is on no OPH output list. It does not by itself fix the baryon density, the neutrino density, the radiation density, the curvature sector, or a conserved homogeneous anomaly charge. A standalone OPH cosmology needs a theorem of the form

$$Q_A \equiv \mathcal{E}_{A,r} = F_{A,r}(Q^{\mathrm{rel}}_{A,r}, \mu^{\mathrm{rel}}_{A,r}, P_\star, N_{\mathrm{CRC}}, \mathfrak{R}^{\mathrm{rel}}_{A,r}, \mathcal{P}_r)$$

or an explicitly declared state-selection premise such as a flat benchmark using (72).

17.15 Source-Only Anomaly Abundance Selector

Definition 17.1 (C.52. Anomaly release state)**.** *An anomaly release state at regulator r is*

$$\mathfrak{R}^{\mathrm{rel}}_{A,r} = (\Sigma^{\mathrm{rel}}_r, n^a_r, a_{\mathrm{rel,r}}, V_{\mathrm{com},r}, \Pi^{(0)}_r, \mathfrak{B}^{\mathrm{rel}}_r, h_{\mathrm{rel}}, \mathfrak{L}_{\mathrm{no\,data}}).$$

Here Σ^{rel}_r is the finite release hypersurface; n^a_r is its future unit normal in the parent tetrad; $a_{\mathrm{rel,r}}$ is the release scale factor; $V_{\mathrm{com},r}$ is the declared comoving volume; $\Pi^{(0)}_r$ is the homogeneous FLRW zero-mode projector; and $\mathfrak{B}^{\mathrm{rel}}_r$ is the release-branch boundary and sector data, including declared $P_\star$, N_{CRC}, compact-gauge branch, ordinary-sector branch, and exchange branch. The hash h_{rel} freezes the generation. The ledger $\mathfrak{L}_{\mathrm{no\,data}}$ records the source dependency graph.

The release state is source-admissible only if all of its fields are fixed before CMB, BAO, supernova, weak-lensing, RSD, SPARC, cluster, and other likelihood comparisons.

Definition 17.2 (C.53. Finite anomaly load observable)**.** *Let*

$$\mathcal{P}_r[X, g]$$

be the finite covariant collar-packet parent, or its source-side pre-load parent before total homogeneous amplitude selection. For a quotient state $q \in Q^{\mathrm{rel}}_{A,r}$, define the local anomaly energy density on the release hypersurface by

$$\varepsilon_{A,r}(c; q) := T^{ab}_{A,r}(c; q) n_a n_b.$$

On the BW source route this readout has the form

$$\varepsilon_{A,r}(c; q) = \frac{15\hbar c}{8\pi^2} \frac{R^{\mathrm{src}}_{A,r}(c, n, \ell; q)}{\ell^4} + \Delta^{\mathrm{BW}}_{A,r}(c; q),$$

where $\Delta^{\mathrm{BW}}_{A,r}$ is the finite-ball, curvature, gradient, localization, tomography, and variational/moment residual package required by the finite parent receipt.

The finite anomaly load observable is

$$\boxed{\mathsf{L}_{A,r}(q) := \Pi_r^{(0)} \sum_{c\subset\Sigma_r^{\mathrm{rel}}(q)} V_{c,r}^{\mathrm{phys}}(q)\,\varepsilon_{A,r}(c;q).}$$

It has units of energy. It is defined on the physical quotient. Hidden representatives are not valid selector inputs.

The abundance selector is circular and invalid if $\mathsf{L}_{A,r}$ is computed from a parent occupation field whose normalization was set by an externally supplied Ω_A, ρ_{A0}, Planck residual, SPARC fit, cluster fit, Boltzmann residual, or likelihood comparison.

Definition 17.3 (C.54. Source-only anomaly abundance selector)**.** *Let $Q_{A,r}^{\mathrm{rel}}$ be the release quotient state space and let $\mu_{A,r}^{\mathrm{rel}}$ be the declared OPH release law on that quotient. The source-only anomaly abundance selector is the deterministic functional*

$$\boxed{F_{A,r}(Q_{A,r}^{\mathrm{rel}}, \mu_{A,r}^{\mathrm{rel}}, P_\star, N_{\mathrm{CRC}}, \mathfrak{R}_{A,r}^{\mathrm{rel}}, \mathcal{P}_r) := \mathbb{E}_{\mu_{A,r}^{\mathrm{rel}}}[\mathsf{L}_{A,r}].}$$

Equivalently,

$$\boxed{\mathcal{E}_{A,r} = \sum_{q\in Q_{A,r}^{\mathrm{rel}}} \mu_{A,r}^{\mathrm{rel}}(q)\,\mathsf{L}_{A,r}(q).}$$

The emitted homogeneous anomaly density is

$$\boxed{\bar\rho_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\mathrm{com},r}}}$$

on the conserved dust branch.

Definition 17.4 (C.55. Source MaxEnt release law)**.** *The release law $\mu_{A,r}^{\mathrm{rel}}$ is source-admissible if it is either*

$$\mu_{A,r}^{\mathrm{rel}} = \mu_r|_{Q_{A,r}^{\mathrm{rel}}}$$

renormalized on the release branch, where μ_r is a certified OPH quotient law, or the finite MaxEnt law

$$\mu_{A,r}^{\mathrm{rel}}(q) = \frac{m_{A,r}^{\mathrm{rel}}(q)\exp[-S_{A,r}^{\mathrm{rel}}(q) - \sum_j \theta_j F_{j,r}^{\mathrm{rel}}(q)]}{Z_{A,r}^{\mathrm{rel}}}$$

on $Q_{A,r}^{\mathrm{rel}}$, where every constraint $F_{j,r}^{\mathrm{rel}}$, target $c_{j,r}$, unit, support, zero-mode treatment, sector convention, and refinement map is declared in the source constraint ledger.

No simulator output, Boltzmann residual, likelihood value, observational target, or tuned visual diagnostic may appear in $m_{A,r}^{\mathrm{rel}}$, $S_{A,r}^{\mathrm{rel}}$, $F_{j,r}^{\mathrm{rel}}$, $c_{j,r}$, or the branch-selection ledger.

Theorem 17.5 (C.56. Source-only anomaly abundance selection)**.** *Assume:*

1. *$Q_{A,r}^{\mathrm{rel}}$, $\mu_{A,r}^{\mathrm{rel}}$, $P_\star$, N_{CRC}, $\mathfrak{R}_{A,r}^{\mathrm{rel}}$, and $\mathcal{P}_r$ are source-only artifacts.*
2. *$\mathsf{L}_{A,r}$ factors through the physical quotient.*
3. *The local BW/source route, tomography, finite parent, stress closure, gauge, and refinement receipts required by the parent pass.*

4. $\mu^{\mathrm{rel}}_{A,r}$ is a certified OPH quotient law or a source MaxEnt law.

5. The no-data-use ledger excludes all observational likelihoods and all diagnostic solver outputs used to evaluate the prediction.

Then

$$\mathcal{E}_{A,r} = F_{A,r}(Q^{\mathrm{rel}}_{A,r}, \mu^{\mathrm{rel}}_{A,r}, P_\star, N_{\mathrm{CRC}}, \mathfrak{R}^{\mathrm{rel}}_{A,r}, \mathcal{P}_r)$$

is a source-only OPH anomaly abundance artifact. If the conserved dust branch holds, then

$$\bar{\rho}_{A,r}(a) = \frac{\mathcal{E}_{A,r}}{a^3 V_{\mathrm{com},r}}$$

is a source-only homogeneous anomaly density.

Proof. The load observable $\mathsf{L}_{A,r}$ is constructed from the parent-emitted anomaly stress or from the BW-normalized source route followed by stress tomography. By assumption it factors through $Q^{\mathrm{rel}}_{A,r}$, so hidden carrier coordinates, port labels, repair schedules, gauge representatives, and worker metadata do not change it.

The release law $\mu^{\mathrm{rel}}_{A,r}$ is either a certified OPH quotient law or the finite MaxEnt law determined by source constraints. Hence the expectation

$$\mathbb{E}_{\mu^{\mathrm{rel}}_{A,r}}[\mathsf{L}_{A,r}]$$

is a deterministic finite functional of source artifacts.

No observational likelihood or diagnostic solver output enters the source DAG by assumption. Therefore the artifact is source-only. On the conserved dust branch, the background parent equation gives

$$(a^3 \bar{\rho}_A V_{\mathrm{com}})' = 0,$$

so the selected release load transports as

$$\bar{\rho}_A(a) = \mathcal{E}_{A,r}/(a^3 V_{\mathrm{com}}).$$

□

Theorem 17.6 (C.57. No-data-use proof for $F_{A,r}$)**.** *Let G^{load}_A be the source DAG whose terminal node is $\mathcal{E}_{A,r}$. Suppose every node verifier recomputes its predicate from committed artifacts, theorem witnesses, parent predicates, dependency manifests, and error envelopes, and suppose every root node is one of*

$$Q^{\mathrm{rel}}_{A,r},\quad m^{\mathrm{rel}}_{A,r},\quad S^{\mathrm{rel}}_{A,r},\quad F^{\mathrm{rel}}_{j,r},\quad c_{j,r},\quad P_\star,\quad N_{\mathrm{CRC}},\quad \mathfrak{R}^{\mathrm{rel}}_{A,r},\quad \mathcal{P}_r$$

or a standard non-CMB physical constant allowed by the cosmology contract.

If the dependency ledger contains no CMB, BAO, supernova, weak-lensing, RSD, SPARC, cluster, likelihood, posterior, nuisance-fit, simulator residual, or diagnostic visual-comparison input, then

`ANOMALY_ABUNDANCE_SOURCE_RECEIPT`

passes.

Proof. Topologically order G^{load}_A. Root nodes are source-only by hypothesis. For a non-root node, the verifier accepts only if every parent has passed and the node value is recomputed from its declared parents and theorem witness. Induction over the DAG proves that the terminal value $\mathcal{E}_{A,r}$ is a deterministic function of source-only inputs. Since the ledger excludes every observational or likelihood target used for evaluation, no target-leakage path exists. A producer Boolean is not a proof because it is not a recomputation from the committed artifacts. □

Theorem 17.7 (C.58. Exact refinement compatibility of the abundance selector)**.** *Let $s \succeq r$ and let*

$$c_{sr} : Q^{\mathrm{rel}}_{A,s} \to Q^{\mathrm{rel}}_{A,r}$$

be the physical coarse map. Suppose:

1. *the release laws are exactly compatible:*

$$(c_{sr})_{\#}\mu^{\mathrm{rel}}_{A,s} = \mu^{\mathrm{rel}}_{A,r};$$

2. *the load observable is exactly natural:*

$$\mathsf{L}_{A,s}(q_s) = \mathsf{L}_{A,r}(c_{sr}q_s) \qquad \forall q_s \in Q^{\mathrm{rel}}_{A,s};$$

3. *the release volume and homogeneous projector commute with refinement.*

Then

$$\boxed{\mathcal{E}_{A,s} = \mathcal{E}_{A,r}.}$$

Proof. By definition,

$$\mathcal{E}_{A,s} = \sum_{q_s} \mu^{\mathrm{rel}}_{A,s}(q_s)\mathsf{L}_{A,s}(q_s).$$

Using load naturality,

$$\mathcal{E}_{A,s} = \sum_{q_s} \mu^{\mathrm{rel}}_{A,s}(q_s)\mathsf{L}_{A,r}(c_{sr}q_s).$$

Push the fine law forward along c_{sr}. Exact law compatibility gives

$$\sum_{q_s : c_{sr}q_s = q_r} \mu^{\mathrm{rel}}_{A,s}(q_s) = \mu^{\mathrm{rel}}_{A,r}(q_r).$$

Therefore

$$\mathcal{E}_{A,s} = \sum_{q_r} \mu^{\mathrm{rel}}_{A,r}(q_r)\mathsf{L}_{A,r}(q_r) = \mathcal{E}_{A,r}.$$

□

Theorem 17.8 (C.59. Approximate refinement bound for the abundance selector)**.** *Let $s \succeq r$. Suppose*

$$\left\|(c_{sr})_{\#}\mu^{\mathrm{rel}}_{A,s} - \mu^{\mathrm{rel}}_{A,r}\right\|_{\mathrm{TV}} \le \delta^{\mu}_{sr},$$

and

$$\sup_{q_s} |\mathsf{L}_{A,s}(q_s) - \mathsf{L}_{A,r}(c_{sr}q_s)| \le \delta^{L}_{sr}.$$

Let

$$L_{\max,r} := \|\mathsf{L}_{A,r}\|_{\infty}.$$

Then

$$\boxed{|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \le \delta^{L}_{sr} + 2L_{\max,r}\delta^{\mu}_{sr}.}$$

For a refinement chain $r = r_0 \preceq r_1 \preceq \cdots \preceq r_n = s$,

$$|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \le \sum_{k=0}^{n-1} \left(\delta^{L}_{r_{k+1}r_k} + 2L_{\max,r_k}\delta^{\mu}_{r_{k+1}r_k}\right).$$

If the right-hand side tends to zero along a cofinal family, the selected anomaly load has a regulator-independent limit.

Proof. Insert and subtract the coarse load pulled back to the fine law:

$$|\mathcal{E}_{A,s} - \mathcal{E}_{A,r}| \le |\mathbb{E}_{\mu_s}[\mathsf{L}_{A,s} - \mathsf{L}_{A,r} \circ c_{sr}]| \\ + |\mathbb{E}_{\mu_s}[\mathsf{L}_{A,r} \circ c_{sr}] - \mathbb{E}_{\mu_r}[\mathsf{L}_{A,r}]| \,.$$

The first term is at most δ^L_{sr}. The second term is bounded by the total-variation estimate

$$2\|\mathsf{L}_{A,r}\|_\infty \delta^\mu_{sr}.$$

Summing the one-step bounds gives the chain estimate. Cofinal convergence follows immediately. □

Theorem 17.9 (C.60. Source-only ρ_A promotion)**.** *The receipt*

$$\mathtt{RHO\_A\_SOURCE\_RECEIPT}$$

passes iff

$$\mathtt{RHO\_A\_TRANSPORT\_RECEIPT} \wedge \mathtt{ANOMALY\_ABUNDANCE\_SOURCE\_RECEIPT}$$

passes under the same frozen generation, unit ledger, release state, and parent generation. When this receipt passes, $\bar{\rho}_A(a)$ may be labelled

$$\mathtt{SOURCE\_ONLY\_ANOMALY\_ABUNDANCE}.$$

If only the transport receipt passes, the label is

$$\mathtt{PHYSICAL\_PARENT\_WITH\_CONDITIONAL\_ABUNDANCE}.$$

Proof. The transport receipt proves that the finite parent evolves a homogeneous anomaly load consistently once that load is supplied. The abundance source receipt proves that the load itself is selected by the OPH source functional $F_{A,r}$, without target leakage and with refinement compatibility. Their conjunction is exactly the statement that ρ_A is both physically transported and source-selected. Without the selector, the amplitude is external state data. □

The closed paper-side abundance receipt is

$$\begin{aligned}\mathtt{ANOMALY\_ABUNDANCE\_SOURCE\_RECEIPT} = {} & \mathtt{ANOMALY\_RELEASE\_STATE\_RECEIPT} \wedge \mathtt{ANOMALY\_LOAD\_OBSERVABLE\_RECEI} \\ & \wedge \mathtt{SOURCE\_MAXENT\_RELEASE\_LAW\_RECEIPT} \wedge \mathtt{LOAD\_QUOTIENT\_INVARIANC} \\ & \wedge \mathtt{LOAD\_NO\_CIRCULARITY\_RECEIPT} \wedge \mathtt{LOAD\_NO\_DATA\_USE\_RECEIPT} \\ & \wedge \mathtt{LOAD\_REFINEMENT\_COMPATIBILITY\_RECEIPT}.\end{aligned}$$

Then

$$\mathtt{RHO\_A\_SOURCE\_RECEIPT} = \mathtt{RHO\_A\_TRANSPORT\_RECEIPT} \wedge \mathtt{ANOMALY\_ABUNDANCE\_SOURCE\_RECEIPT}.$$

The source-ready dark Boltzmann handoff is

$$\begin{aligned}\mathtt{DARK\_PHYSICAL\_BOLTZMANN\_SOURCE\_RECEIPT} = {} & \mathtt{FINITE\_COVARIANT\_PARENT\_RECEIPT\_V2} \wedge \mathtt{PHYSICAL\_ANOMALY\_K} \\ & \wedge \mathtt{RHO\_A\_TRANSPORT\_RECEIPT} \wedge \mathtt{FROZEN\_GENERATION\_MANIFEST\_} \\ & \wedge \mathtt{BOLTZMANN\_INITIAL\_CONSTRAINT\_RECEIPT} \wedge \mathtt{BOLTZMANN\_ADAP} \\ & \wedge \mathtt{CDM\_LIMIT\_RECOVERY\_RECEIPT}.\end{aligned}$$

For a source-only dark abundance claim,

$$\texttt{SOURCE\_ONLY\_ANOMALY\_ABUNDANCE} \Longleftrightarrow \texttt{RHO\_A\_SOURCE\_RECEIPT}.$$

Equivalently, the selector computes

$$\boxed{F_{A,r} = \mathbb{E}_{\mu^{\mathrm{rel}}_{A,r}}\left[\Pi^{(0)}_r \sum_{c\subset\Sigma^{\mathrm{rel}}_r} V^{\mathrm{phys}}_{c,r}\, T^{ab}_{A,r}(c) n_a n_b\right].}$$

On the BW source branch,

$$\boxed{F_{A,r} = \mathbb{E}_{\mu^{\mathrm{rel}}_{A,r}}\left[\Pi^{(0)}_r \sum_{c\subset\Sigma^{\mathrm{rel}}_r} V^{\mathrm{phys}}_{c,r}\left(\frac{15\hbar c}{8\pi^2}\frac{R^{\mathrm{src}}_{A,r}(c,n,\ell)}{\ell^4} + \Delta^{\mathrm{BW}}_{A,r}(c)\right)\right].}$$

The selector integrates the finite source stress over the release hypersurface and averages it over the declared source quotient ensemble. It does not read a Friedmann residual, a Planck fit, a SPARC fit, a cluster fit, or a Boltzmann likelihood.

18 Falsifiers

The OPH dark branch fails if any of the following happens:

1. OPH collar/cut microphysics cannot derive the activation law $p(x) = 1 - \exp[-\lambda_{\mathrm{collar}}\sqrt{x}]$, or a replacement law that passes the same galaxy tests.
2. The carried galaxy defect is not positive or not attractive.
3. The settled anomaly stress cannot give $\Phi = \Psi$.
4. A full SPARC likelihood rejects both the unit branch and every independently derived coefficient branch under the same nuisance treatment used for comparison models.
5. Cluster lensing offsets require a relaxation law incompatible with old settled-galaxy RAR.
6. CMB peaks, CMB lensing, BAO, weak lensing, or growth reject every frozen finite covariant parent whose certified Γ_{rec}, $B_A(k,a)$, $\rho_A(a)$, stress closure, recipient/exchange-current channel, physical clock, and CDM-limit recovery are compatible with the galaxy branch.

19 Reproducibility

The numerical tables were produced in the OPH cosmology workspace by:

```
python3 code/dark_matter/scripts/dark_sector_measurement_ledger.py
python3 code/dark_matter/scripts/dark_empirical_scorecard.py --quiet
python3 code/dark_matter/scripts/sparc_systematic_likelihood.py
python3 code/dark_matter/scripts/dark_perturbation_parameters.py
python3 code/dark_matter/scripts/dark_repair_transition_matrix.py
python3 code/dark_matter/scripts/dark_cluster_boltzmann_prelikelihood.py
```

```
python3 code/dark_matter/scripts/dark_jacobi_repair_clock.py
python3 code/dark_matter/scripts/dark_homogeneous_state_selection.py
python3 code/dark_matter/scripts/dark_parent_collar_grid.py --diagnostic-scalar-load \
  --mu-eq 5.363470440729118 \
  --out code/dark_matter/outputs/dark_parent_flat_capacity_state_selection.json
python3 code/dark_matter/scripts/dark_cmb_bao_growth_s8_likelihood.py \
  --parent-grid code/dark_matter/outputs/dark_parent_flat_capacity_state_selection.json
python3 code/dark_matter/scripts/camb_fixed_neutrino_compare.py
python3 -m oph_fpe.cli dark-sector-simulation-plan --run-dir runs/<run_id> \
  --out runs/<run_id>/dark_sector_simulation_plan
```

The SPARC data files used are public measurement tables from the SPARC page [7].

References

[1] B. Mueller, A. Osika, M. Poneder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Recovering Relativity and the Standard Model from Observer Overlap Consistency*, OPH compact paper, GitHub main PDF. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_relativity_and_standard_model_structure_from_observer_overlap_consistency_compact.pdf

[2] B. Mueller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, M. A. Visser, K. A. Anirudha, D. Matscheko, and J. Hill, *Observers Are All You Need*, OPH synthesis paper, GitHub main PDF. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/observers_are_all_you_need.pdf

[3] B. Mueller, A. Osika, K. Xue, B. Cassie, M. A. Visser, and D. Matscheko, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*, OPH microphysics paper, GitHub main PDF. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf

[4] B. Mueller, A. Osika, M. Poneder, K. Xue, M. A. Visser, and D. Matscheko, *Deriving the Particle Zoo from Observer Consistency*, OPH particle paper, GitHub main PDF. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_the_particle_zoo_from_observer_consistency.pdf

[5] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, arXiv:1807.06209. https://arxiv.org/abs/1807.06209

[6] F. Lelli, S. S. McGaugh, and J. M. Schombert, *SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves*, Astron. J. 152, 157, 2016, arXiv:1606.09251. https://arxiv.org/abs/1606.09251

[7] F. Lelli, S. S. McGaugh, and J. M. Schombert, SPARC public data page. https://astroweb.cwru.edu/SPARC/

[8] S. S. McGaugh, F. Lelli, and J. M. Schombert, *The Radial Acceleration Relation in Rotationally Supported Galaxies*, Phys. Rev. Lett. 117, 201101, 2016, arXiv:1609.05917. https://arxiv.org/abs/1609.05917

[9] F. Lelli, S. S. McGaugh, and J. M. Schombert, *The Small Scatter of the Baryonic Tully–Fisher Relation*, Mon. Not. R. Astron. Soc. 484, 3267–3278, 2019, arXiv:1901.05966. https://arxiv.org/abs/1901.05966

[10] H. Desmond, *The underlying radial acceleration relation*, Mon. Not. R. Astron. Soc. 526, 3342–3356, 2023, arXiv:2303.11314. https://arxiv.org/abs/2303.11314

[11] R. S. Park et al., *Cassini spacecraft navigation data constrain modified gravity in the Solar System*, 2026, arXiv:2602.17884. https://arxiv.org/abs/2602.17884

[12] Gaia Collaboration, *Gaia Early Data Release 3: Acceleration of the Solar System from Gaia astrometry*, Astron. Astrophys. 649, A9, 2021, arXiv:2012.02036. https://arxiv.org/abs/2012.02036

[13] P. Li, F. Lelli, S. S. McGaugh, and J. M. Schombert, *Fitting the radial acceleration relation to individual SPARC galaxies*, Astron. Astrophys. 615, A3, 2018, arXiv:1803.00022. https://arxiv.org/abs/1803.00022

[14] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. 270, 365, 1983. https://ui.adsabs.harvard.edu/abs/1983ApJ...270..365M/abstract

[15] J. D. Bekenstein, *Relativistic gravitation theory for the modified Newtonian dynamics paradigm*, Phys. Rev. D 70, 083509, 2004. https://doi.org/10.1103/PhysRevD.70.083509

[16] D. Clowe, M. Bradac, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky, *A direct empirical proof of the existence of dark matter*, Astrophys. J. Lett. 648, L109, 2006, arXiv:astro-ph/0608407. https://arxiv.org/abs/astro-ph/0608407